Lecture 9: Asymptotic Theory of Testing

SDS8102

Lecture 9

Overview. Two tests may both detect a fixed departure from the null when the sample is large, yet require very different amounts of data to reach the same power. Consistency alone does not explain this difference. We therefore let the departure shrink as the sample size grows, so that detecting it remains difficult. In regular models, a departure of order $n^{-1/2}$ stays comparable to sampling noise. On this scale, the mean and sign tests provide a first comparison: their limiting power curves reveal how efficiently they detect a small location shift and how that efficiency translates into required sample size.

This comparison leads to a broader question: how much local power can any test achieve? To answer it, we turn from individual statistics to the likelihood. Local asymptotic normality (LAN) gives a Gaussian model for the local likelihood, and the asymptotic representation theorem transfers its testing bound to the original problem. The same quadratic likelihood expansion then explains why Wald, Rao, and likelihood-ratio tests have the same first-order local power under regularity conditions. The chapter thus moves from comparing particular tests to identifying the information bound on their performance and the procedures that attain it.

Prerequisites. The law of large numbers (LLN) and central limit theorem (CLT) for independent, identically distributed (iid) observations; convergence in probability and distribution; Taylor expansion; and asymptotic normality of estimators. Section 0 recalls the notation, influence functions, sandwich covariance, and maximum likelihood estimator (MLE) expansion used below.

Reading. van der Vaart (1998): §§14.1–14.3 for local power and relative efficiency; Chapter 6 for contiguity; §§7.1–7.3 for LAN; §§15.1–15.3 for representation and optimality, with §9.2 giving the broader representation result; Chapter 16 for likelihood tests. Lehmann and Romano (2005), third edition: §§12.2–12.3 for LAN; §§12.4.2–12.4.4 for Wald, Rao, and likelihood-ratio tests; §§13.3–13.5 for further asymptotic optimality. The References section gives full bibliographic details.

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0 Notation and prerequisites

Consider iid observations $X_1, \dots, X_n \sim P_\theta$ in a sample space $\mathcal{X}$, with $\theta \in \Theta \subset \mathbb{R}^d$ and an interior reference value θ_0. Scalar examples have $d = 1$. Define

$$P_n f = \frac{1}{n}\sum_{i=1}^{n} f(X_i), \qquad \theta_n(h) = \theta_0 + \frac{h}{\sqrt{n}}, \qquad P_{n,h} = P_{\theta_n(h)}^{\otimes n}.$$

Here $P_n f$ is an empirical average, and $P_\theta^{\otimes n}$ is the joint law of n independent observations with common law P_θ. The local vector h is fixed; all statements concern sufficiently large n for which $\theta_n(h) \in \Theta$. We write $\mathbb{E}_0$ for expectation under P_{θ_0} and $\mathbb{E}_{n,h}$ for expectation under $P_{n,h}$.

We write $Y_n \rightsquigarrow Y$ for *convergence in distribution*, also called *weak convergence*. For random vectors in $\mathbb{R}^k$, this means

$$\mathbb{E}f(Y_n) \longrightarrow \mathbb{E}f(Y) \quad \text{for every bounded continuous } f : \mathbb{R}^k \to \mathbb{R}.$$

Each expectation uses the law of its argument; the random vectors need not share a probability space. Under $P_{n,h}$, the left side is $\mathbb{E}_{n,h} f(Y_n)$. The notation $N_k(m, \Sigma)$ denotes a k-dimensional normal law with mean m and covariance Σ; a law on the right of $\rightsquigarrow$ specifies the limiting distribution. We reserve $\xrightarrow{\mathbb{P}}$ for convergence in probability: $Y_n \xrightarrow{\mathbb{P}} y$ means $\mathbb{P}(\|Y_n - y\| > \epsilon) \to 0$ for every $\epsilon > 0$.

Write $Y_n = o_{\mathbb{P}}(1)$ for $Y_n \xrightarrow{\mathbb{P}} 0$. Write $Y_n = O_{\mathbb{P}}(1)$ for boundedness in probability:

$$\lim_{M\to\infty} \limsup_{n\to\infty} \mathbb{P}(\|Y_n\| > M) = 0.$$

For deterministic $a_n > 0$, $Y_n = o_{\mathbb{P}}(a_n)$ or $O_{\mathbb{P}}(a_n)$ means $Y_n/a_n = o_{\mathbb{P}}(1)$ or $O_{\mathbb{P}}(1)$, respectively. These statements use the law specified at that point. Vector norms are Euclidean; matrix norms are induced operator norms, $\|C\| = \sup_{\|u\|=1} \|Cu\|$. Superscript T denotes transpose, $A^{-T} = (A^{-1})^T$, and I_r is the $r \times r$ identity matrix. Throughout, $0 < \alpha < 1$, Φ is the standard normal cumulative distribution function (cdf), and $z_q = \Phi^{-1}(q)$ for $0 < q < 1$.

For a vector estimating function $\psi_\theta : \mathcal{X} \to \mathbb{R}^d$, write

$$\Psi_n(\theta) = P_n\psi_\theta, \quad \Psi(\theta) = \mathbb{E}_0\psi_\theta(X), \quad A = \dot{\Psi}(\theta_0), \quad B = \mathrm{Var}_0(\psi_{\theta_0}(X)).$$

Here $\dot{\Psi}(\theta_0)$ is the derivative matrix at θ_0. For a consistent root of $\Psi_n(\widehat{\theta}_n) = 0$, the asymptotic linear representation used below is

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varphi(X_i) + o_{\mathbb{P}}(1), \qquad \varphi = -A^{-1}\psi_{\theta_0}. \tag{0.1}$$

The function φ is the influence function: one observation's first-order contribution to the estimator. Whenever we invoke (0.1), we assume nonsingular A, $\mathbb{E}_0\varphi = 0$, and finite second moments. Its covariance $V = \mathbb{E}_0\varphi\varphi^T = A^{-1}BA^{-T}$ is the sandwich covariance. Section 5 derives the likelihood version under explicit smoothness conditions.

The iid law of large numbers and multivariate central limit theorem are prerequisites. Appendix G supplies the shared limit and matrix calculations. The Portmanteau lemma is stated without proof in Section 1.3.

1 Local power of statistical tests

A test should control false rejections and detect departures from the null. At a fixed alternative, however, many different tests eventually have power one. To compare the observations they need, we let the alternative approach the null at the scale of estimation error. The resulting local power depends on a single standardized slope. We first derive that dependence and calculate the slope for two location tests. Here and in Section 2, the parameter and local shift are scalar ($d = 1$). Section 3 returns to the general vector model.

Remark 1.1 (Intuition). An estimator fluctuates like an average of independent contributions. Testing compares a displacement from the null with those fluctuations. To calculate local power, the required expansion must hold under the local sample law, rather than only at a fixed parameter.

1.1 Test functions and error control

A test function is a measurable map $\phi_n : \mathcal{X}^n \to [0, 1]$. Its value is the probability of rejection given the observed data. Write $\mathbf{1}\{A\}$ for the indicator of an event A, equal to one on A and zero otherwise. A rule with values in $\{0, 1\}$ needs no randomization. If U is independent uniform on $[0, 1]$, the decision $\mathbf{1}\{U \le \phi_n\}$ implements a general test function and has rejection probability $\mathbb{E}_\theta \phi_n$.

Write $\beta_n(\theta) = \mathbb{E}_\theta \phi_n$ for the power, with expectation here taken over the sample. The disjoint sets $\Theta_0, \Theta_1 \subseteq \Theta$ specify the null and alternative hypotheses. For testing $\theta \in \Theta_0$ against $\theta \in \Theta_1$, the size is $\sup_{\theta \in \Theta_0} \beta_n(\theta)$. Level α means that this size is at most α.

Definition 1.2 (Two asymptotic level conditions)**.** Uniform asymptotic level α means

$$\limsup_{n \to \infty} \sup_{\theta \in \Theta_0} \beta_n(\theta) \le \alpha.$$

Pointwise asymptotic level α means

$$\limsup_{n \to \infty} \beta_n(\theta) \le \alpha \quad \text{for each fixed } \theta \in \Theta_0.$$

For a simple null, these conditions coincide.

A worst-case null parameter can move with n, so pointwise control need not imply uniform control.

Definition 1.3 (Consistency)**.** A sequence of tests is consistent against Θ_1 if $\beta_n(\theta) \to 1$ for every fixed $\theta \in \Theta_1$.

Local power distinguishes consistent tests by their ability to detect departures that shrink with the sample size.

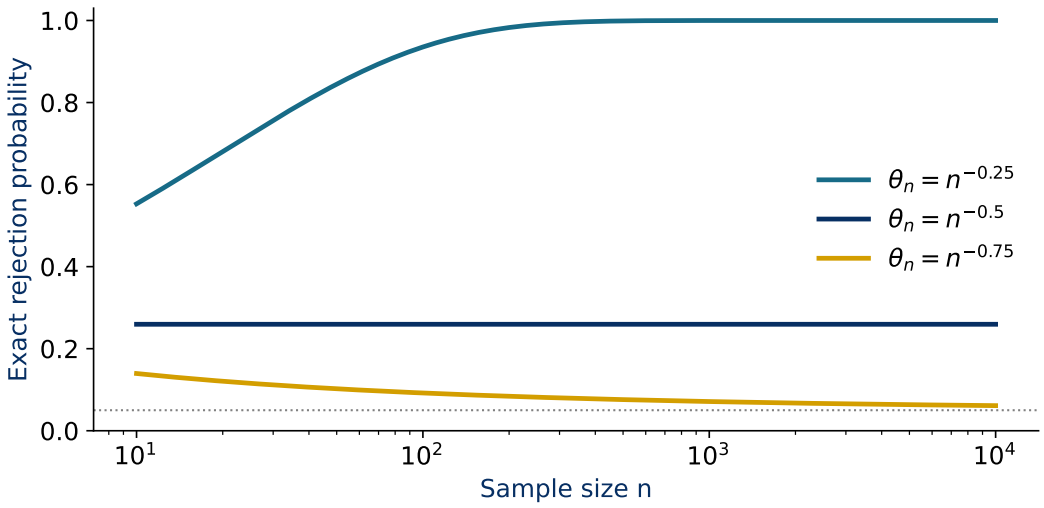


Figure 1: Exact normal rejection probabilities at level .05, $\sigma = 1$, and $\theta_n = n^{-\gamma}$. The root-n rate gives nontrivial limiting power.

1.2 Local alternatives and the root-n scale

Example 1.4 (Normal mean, known variance)**.** Let $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, with known $\sigma > 0$, and write $\overline{X}_n = n^{-1}\sum_i X_i$ for the sample mean. To test $\theta \le 0$ against $\theta > 0$, reject when $Z_n = \sqrt{n}\overline{X}_n/\sigma > z_{1-\alpha}$. Since $Z_n - \sqrt{n}\theta/\sigma$ is exactly standard normal,

$$\beta_n(\theta) = 1 - \Phi\left(z_{1-\alpha} - \frac{\sqrt{n}\theta}{\sigma}\right). \tag{1.1}$$

This increases in θ and equals α at zero, proving exact level over $\theta \le 0$. It tends to one at every fixed $\theta > 0$.

For $\theta_n = hn^{-\gamma}$ with $h > 0$ and $\gamma > 0$, the standardized signal is $hn^{1/2-\gamma}/\sigma$. Formula (1.1) gives limiting power one if $\gamma < 1/2$, level α if $\gamma > 1/2$, and $1 - \Phi(z_{1-\alpha} - h/\sigma)$ if $\gamma = 1/2$. Only the last scale retains a nontrivial comparison.

Definition 1.5 (Local asymptotic power)**.** For the local laws $P_{n,h}$ in Section 0, the local asymptotic power function is

$$\beta(h) = \lim_{n\to\infty} \mathbb{E}_{n,h}\phi_n,$$

when the limit exists. The local parameter h is fixed in this limit.

The root-n scale reflects regular finite-dimensional estimation. Other models can have different detection rates.

1.3 Power from a local normal limit

A normal approximation gives convergence in distribution. The following standard Portmanteau lemma links this convergence to probabilities of sets; see van der Vaart [1, Lemma 2.2]. We state it without proof. Borel sets are the measurable sets generated by the open sets of Euclidean space;

$\overline{D}$ and D° denote closure and interior.

Lemma 1.6 (Portmanteau). *Let X_n and X be random vectors in $\mathbb{R}^k$, and suppose $X_n \rightsquigarrow X$. The vectors X_n may be defined on different probability spaces, so their underlying probability laws may change with n. Then:*

(i) For every closed set $F \subseteq \mathbb{R}^k$,

$$\limsup_{n\to\infty} \mathbb{P}(X_n \in F) \le \mathbb{P}(X \in F).$$

(ii) For every Borel set $D \subseteq \mathbb{R}^k$ with $\mathbb{P}(X \in \partial D) = 0$,

$$\mathbb{P}(X_n \in D) \longrightarrow \mathbb{P}(X \in D),$$

where $\partial D = \overline{D} \setminus D^\circ$ is the boundary of D.

Theorem 1.7 (Normal local power). *Fix $\kappa > 0$ and $\alpha \in (0,1)$. For every fixed h, suppose under $P_{n,h}$ that*

$$T_n \rightsquigarrow N(\kappa h, 1), \qquad c_n \xrightarrow{\mathbb{P}} z_{1-\alpha}.$$

Then $\phi_n = \mathbf{1}\{T_n > c_n\}$ has limiting power

$$\mathbb{E}_{n,h}\phi_n \to 1 - \Phi(z_{1-\alpha} - \kappa h). \tag{1.2}$$

For $\mathbf{1}\{|T_n| > c_n^{(2)}\}$, where $c_n^{(2)} \xrightarrow{\mathbb{P}} z_{1-\alpha/2}$ under the same laws, the limiting power is

$$1 - \Phi(z_{1-\alpha/2} - \kappa h) + \Phi(-z_{1-\alpha/2} - \kappa h). \tag{1.3}$$

The coefficient κ in the limiting mean κh is the *local slope* of the standardized statistic: it measures the change in its limiting mean per unit local displacement h. The theorem converts that shift into a rejection probability.

Proof of Theorem 1.7. Fix h and work under $P_{n,h}$. Set $c = z_{1-\alpha}$, and let $Z = \kappa h + Z_0$ with $Z_0 \sim N(0,1)$. By Lemma G.1, aka the continuous mapping theorem,

$$T_n - c_n \rightsquigarrow Z - c \qquad \text{under } P_{n,h}.$$

Apply Lemma 1.6(ii) with $X_n = T_n - c_n$, $X = Z - c$, and $D = (0,\infty)$. Here $\partial D = \{0\}$, and $\mathbb{P}(X = 0) = \mathbb{P}(Z = c) = 0$ because Z has a density. Thus

$$\mathbb{E}_{n,h}\phi_n = P_{n,h}(T_n - c_n > 0) \longrightarrow \mathbb{P}(Z - c > 0) = 1 - \Phi(z_{1-\alpha} - \kappa h),$$

which proves (1.2).

For the two-sided rule, set $a = z_{1-\alpha/2} > 0$. By Lemma G.1,

$$|T_n| - c_n^{(2)} \rightsquigarrow |Z| - a \qquad \text{under } P_{n,h}.$$

Apply Lemma 1.6(ii) with $X_n = |T_n| - c_n^{(2)}$, $X = |Z| - a$, and $D = (0,\infty)$. The boundary condition holds since $\mathbb{P}(X = 0) = \mathbb{P}(Z = a) + \mathbb{P}(Z = -a) = 0$. Consequently,

$$P_{n,h}(|T_n| > c_n^{(2)}) \longrightarrow \mathbb{P}(|Z| > a).$$

The events $Z > a$ and $Z < -a$ are disjoint, with probabilities $1 - \Phi(a - \kappa h)$ and $\Phi(-a - \kappa h)$, respectively. Adding them proves (1.3). □

Remark 1.8 (Intuition). Local alternatives shift the standardized mean by κh while the limiting variance stays one. A larger κ moves more probability into the rejection region. The absence of boundary mass is what turns this distributional statement into a statement about power.

Studentization means dividing an estimator's displacement by an estimated standard error. If $\sqrt{n}(\widehat{\theta}_n - \theta_n(h))$ has limiting variance V, this standard error is estimated by $\sqrt{\widehat{V}_n/n}$. The next result applies the normal-power calculation to this estimated scale.

Corollary 1.9 (Studentized estimator test)**.** *Suppose, for each fixed h, under $P_{n,h}$ that*

$$\sqrt{n}(\widehat{\theta}_n - \theta_n(h)) \rightsquigarrow N(0, V), \qquad \widehat{V}_n \xrightarrow{\mathbb{P}} V \in (0, \infty).$$

Then $T_n = \sqrt{n}(\widehat{\theta}_n - \theta_0)/\sqrt{\widehat{V}_n}$ satisfies Theorem 1.7 with $\kappa = 1/\sqrt{V}$. Define T_n arbitrarily when $\widehat{V}_n \le 0$.

Proof of Corollary 1.9. Put $Z_n = \sqrt{n}(\widehat{\theta}_n - \theta_n(h))$. By Lemma G.1, $Z_n = O_{\mathbb{P}}(1)$. Also, $\sqrt{n}(\widehat{\theta}_n - \theta_0) = Z_n + h$. Consistency gives $P_{n,h}(\widehat{V}_n \ge V/2) \to 1$.

Define $g(v) = (\max\{v, V/2\})^{-1/2}$, which is continuous on $\mathbb{R}$ and satisfies $g(V) = V^{-1/2}$. By Lemma G.1, applied to convergence in probability,

$$g(\widehat{V}_n) - V^{-1/2} = o_{\mathbb{P}}(1).$$

When $\widehat{V}_n \ge V/2$, $g(\widehat{V}_n) = \widehat{V}_n^{-1/2}$ and $T_n = (Z_n + h)g(\widehat{V}_n)$. By Lemma G.1,

$$\begin{aligned}(Z_n + h)g(\widehat{V}_n) &= \frac{Z_n + h}{\sqrt{V}} + (Z_n + h)\{g(\widehat{V}_n) - V^{-1/2}\} \\ &= \frac{Z_n + h}{\sqrt{V}} + o_{\mathbb{P}}(1) \rightsquigarrow N(h/\sqrt{V}, 1).\end{aligned}$$

Finally, for every $\epsilon > 0$,

$$P_{n,h}\big(|T_n - (Z_n + h)g(\widehat{V}_n)| > \epsilon\big) \le P_{n,h}(\widehat{V}_n < V/2) \longrightarrow 0.$$

Hence, by Lemma G.1,

$$T_n = (Z_n + h)g(\widehat{V}_n) + o_{\mathbb{P}}(1) \rightsquigarrow N(h/\sqrt{V}, 1).$$

□

Remark 1.10 (The remaining local-distribution question). The corollary assumes a centered normal limit under moving alternatives. A fixed-parameter CLT alone does not establish it. We verify it directly for the next location examples. Section 3 supplies a likelihood argument for general regular models.

1.4 Location tests

The mean test. Let $X_i = \theta + \varepsilon_i$, where the iid errors have mean zero and variance $\sigma^2 \in (0, \infty)$. Set $\theta_0 = 0$ and $\widehat{\sigma}_n^2 = n^{-1} \sum_i (X_i - \overline{X}_n)^2$. A translation leaves this variance unchanged and increases the numerator of an upper-tail mean statistic. Thus the largest null rejection probability over $\theta \leq 0$ occurs at zero. The local limit below gives its asymptotic calibration.

Proposition 1.11 (Local limit of the mean test)**.** *For every fixed h, under $\theta_n = h/\sqrt{n}$,*

$$T_n^{\text{mean}} = \frac{\sqrt{n}\overline{X}_n}{\widehat{\sigma}_n} \rightsquigarrow N(h/\sigma, 1).$$

Define the statistic to be a fixed constant when $\widehat{\sigma}_n = 0$. Its local slope is $\kappa_{\text{mean}} = 1/\sigma$.

Proof of Proposition 1.11. For every n, $X_i - \theta_n$ has the same law as ε_i. Write $\overline{\varepsilon}_n = n^{-1} \sum_i \varepsilon_i$. Therefore

$$\sqrt{n}\overline{X}_n = h + \frac{1}{\sqrt{n}} \sum_i \varepsilon_i, \qquad \widehat{\sigma}_n^2 = \frac{1}{n} \sum_i \varepsilon_i^2 - \overline{\varepsilon}_n^2.$$

The iid CLT gives $n^{-1/2} \sum_i \varepsilon_i \rightsquigarrow N(0, \sigma^2)$. The LLN applies to ε_i^2 because $\mathbb{E}\varepsilon_i^2 < \infty$, and to ε_i because its first absolute moment is finite. Thus $\widehat{\sigma}_n^2 \xrightarrow{\mathbb{P}} \sigma^2$. In particular, the probability of zero sample variance tends to zero. Corollary 1.9, with $\widehat{\theta}_n = \overline{X}_n$ and $V = \sigma^2$, proves the assertion. Replacing n by $n - 1$ in the sample variance multiplies it by $n/(n-1) \to 1$ and gives the same limit. □

The sign test. Now assume separately that the error cdf F is continuous, symmetric about zero, and differentiable at zero with derivative $f(0) > 0$. No moment assumption is needed. Define

$$K_n = \sum_{i=1}^{n} \mathbf{1}\{X_i > 0\}, \qquad T_n^{\text{sign}} = \frac{2(K_n - n/2)}{\sqrt{n}}.$$

At $\theta = 0$, $K_n \sim \text{Bin}(n, 1/2)$.

Proposition 1.12 (Local limit of the sign test)**.** *For each fixed h, under $\theta_n = h/\sqrt{n}$,*

$$T_n^{\text{sign}} \rightsquigarrow N(2f(0)h, 1).$$

Its local slope is $\kappa_{\text{sign}} = 2f(0)$, so upper-tail rejection above $z_{1-\alpha}$ has limiting power $1 - \Phi(z_{1-\alpha} - 2f(0)h)$.

Proof of Proposition 1.12. Symmetry and continuity imply $1 - F(-t) = F(t)$ and $F(0) = 1/2$. Thus the indicators in row n are independent Bernoulli variables with common success probability

$$p_n = P_{n,h}(X_i > 0) = F(h/\sqrt{n}) = \tfrac{1}{2} + \frac{f(0)h}{\sqrt{n}} + o(n^{-1/2}).$$

In particular, $p_n \to 1/2 \in (0, 1)$. Lemma G.3, whose characteristic-function proof allows the parameter to change with n, gives

$$Z_n = \frac{K_n - np_n}{\sqrt{np_n(1 - p_n)}} \rightsquigarrow N(0, 1).$$

The exact decomposition is

$$T_n^{\text{sign}} = 2\sqrt{p_n(1-p_n)}Z_n + 2\sqrt{n}(p_n - 1/2).$$

The first multiplier tends to one, while the second term tends to $2f(0)h$. Lemma G.1 therefore gives the asserted normal limit. The power conclusion follows from Theorem 1.7. □

Remark 1.13 (Intuition). A small positive shift makes some observations cross zero. The available probability mass is determined by $f(0)$. The mean test instead scales its signal by the standard deviation σ. These different slopes will determine the sample-size comparison.

A finite-sample comparison. For $n = 100$, normal errors of variance one, level .05, and $\theta = .25$, the known-variance mean test has exact power 0.803765. A sign test that rejects above 58 successes and with probability 0.2551091 at 58 has exact power 0.631545; its local normal approximation gives 0.636777. Appendix A derives the exact calibration and proves that boundary randomization does not alter the local limit.

Both tests now have a local power function of the same form. The next section translates their slopes into the amount of data needed for a common testing objective.

2 Asymptotic relative efficiency

The preceding section showed that the mean and sign tests have normal local power with different slopes. A larger slope gives greater power at a fixed local displacement. We now ask the equivalent design question: how many observations does each test need for the same level, target power, and small parameter change? Pitman asymptotic relative efficiency (ARE) expresses this comparison as a sample-size ratio.

2.1 The sample-size convention

Fix a level α and a target power $1-\eta \in (\alpha, 1)$. Compare two tests A and B at the same alternatives $\theta_0+\delta$, with $\delta \downarrow 0$. Their sample sizes $n_j(\delta)$ are called calibrated if their powers satisfy $\beta^j_{n_j(\delta)}(\theta_0+\delta) \to 1-\eta$ for $j \in \{A, B\}$, where β^j_n is the power function of test j.

Definition 2.1 (Pitman ARE)**.** Our convention is

$$\text{ARE}(A, B) = \lim_{\delta \downarrow 0} \frac{n_B(\delta)}{n_A(\delta)},$$

when the comparison is well defined and the limit exists.

An ARE greater than one favors A. An ARE less than one means A needs the reciprocal sample-size multiplier. It is important to state this orientation because a reciprocal is easy to report by mistake.

To see how the local slope determines the required sample size, consider a test with positive local slope κ. Using its one-sided local normal limit as a power approximation for a small shift $\delta > 0$ gives

$$\beta_n(\theta_0 + \delta) \approx 1 - \Phi(z_{1-\alpha} - \kappa\sqrt{n}\delta).$$

Here the normal approximation concerns the test statistic; the observations need not be normally distributed. Setting this approximation equal to the target power $1-\eta$ gives $\kappa\sqrt{n}\delta \approx C$, where $C = z_{1-\alpha} + z_{1-\eta} > 0$. Thus

$$n \approx \frac{(z_{1-\alpha} + z_{1-\eta})^2}{\kappa^2\delta^2} = \frac{C^2}{\kappa^2\delta^2}. \tag{2.1}$$

For the same level, target power, and shift, the factor C^2/δ^2 cancels in a sample-size ratio. This calculation suggests that sample sizes vary inversely with the squared local slopes. The following theorem gives the asymptotic statement for calibrated sample-size sequences.

Theorem 2.2 (Squared local slopes)**.** *Suppose test $j \in \{A, B\}$ has one-sided normal local power with positive slope κ_j. Assume there are sample-size sequences $n_j(\delta) \to \infty$ such that*

1. *$\sqrt{n_j(\delta)}\delta \to h_j \in (0, \infty)$;*
2. *the power along those sequences converges to $1 - \Phi(z_{1-\alpha} - \kappa_j h_j)$;*
3. *that limiting power equals $1-\eta$ for both tests.*

Then

$$\mathrm{ARE}(A, B) = \kappa_A^2/\kappa_B^2. \tag{2.2}$$

Proof of Theorem 2.2. For each j, the power assumptions give

$$1 - \Phi(z_{1-\alpha} - \kappa_j h_j) = 1 - \eta, \qquad \Phi(z_{1-\alpha} - \kappa_j h_j) = \eta.$$

Strict monotonicity of Φ permits inversion. Symmetry gives $z_\eta = -z_{1-\eta}$, so

$$\kappa_j h_j = z_{1-\alpha} + z_{1-\eta} =: C.$$

The requirement $1 - \eta > \alpha$ implies $z_{1-\eta} > z_\alpha = -z_{1-\alpha}$, hence $C > 0$. Squaring the assumed convergence $\sqrt{n_j(\delta)}\delta \to h_j = C/\kappa_j$ gives $n_j(\delta)\delta^2 \to C^2/\kappa_j^2$, a positive finite limit. Therefore

$$\frac{n_B(\delta)}{n_A(\delta)} = \frac{n_B(\delta)\delta^2}{n_A(\delta)\delta^2} \longrightarrow \frac{C^2/\kappa_B^2}{C^2/\kappa_A^2} = \frac{\kappa_A^2}{\kappa_B^2}.$$

□

Remark 2.3. Theorem 2.2 compares sample-size sequences whose powers approach a common target. To use the same ratio for the smallest sample sizes attaining that target, we must justify searching over n. During this search, the local displacement $\sqrt{n}\delta$ changes, so the power approximation must hold uniformly over bounded ranges of that displacement. Appendix H makes this argument precise and gives conditions under which the minimum sample sizes have the same limiting ratio.

2.2 Sign versus mean

Under the common symmetric location model, assume also $\sigma^2 < \infty$. Propositions 1.11 and 1.12 give

$$\mathrm{ARE}(\mathrm{sign}, \mathrm{mean}) = 4\sigma^2 f(0)^2. \tag{2.3}$$

Both procedures test the same location parameter in this model. In an unrestricted asymmetric model the mean and median can differ; a sign test and a mean test can then address different hypotheses.

The error shape enters through $\sigma f(0)$. A small shift δ moves approximately $f(0)\delta$ of the probability across zero, whereas the mean has noise scale $\sigma/\sqrt{n}$. Thus the sign test is more efficient exactly when $2\sigma f(0) > 1$. Rescaling the error alone leaves $\sigma f(0)$, and hence the ARE, unchanged.

Example 2.4 (Normal errors)**.** For $N(0, \sigma^2)$ errors, $f(0) = 1/(\sigma\sqrt{2\pi})$. The last two rows below use one-sided level $\alpha = .05$, target power .80, $\sigma = 1$, and $\delta = .25$ in formula (2.1).

Quantity	Mean test	Sign test
Local slope κ	$1/\sigma$	$\sqrt{2/\pi}/\sigma$
$\mathrm{ARE}(\text{test}, \text{mean})$	1	$2/\pi \approx .63662$
Asymptotic sample-size ratio $n_{\text{test}}/n_{\text{mean}}$	1	$\pi/2 \approx 1.5708$
Planning sample size	98.92	155.38
Rounded up	99	156

The sign test needs about 57% more observations because it discards magnitude information. The mean-test calculation is exact for known σ; for the studentized mean and sign tests, the sample sizes are asymptotic planning values.

Example 2.5 (Laplace errors)**.** Laplace errors have density $f(x) = (2b)^{-1}e^{-|x|/b}$ for $x \in \mathbb{R}$, with scale $b > 0$. Then $f(0) = 1/(2b)$ and symmetry gives mean zero. Substitution $u = x/b$ gives

$$\sigma^2 = \frac{1}{b}\int_0^\infty x^2 e^{-x/b}\,dx = b^2\int_0^\infty u^2 e^{-u}\,du = 2b^2,$$

where the last integral equals two by two integrations by parts. Formula (2.3) gives ARE 2. The sign test needs half as many observations as the mean test in this local comparison. Here $\sigma f(0) = 1/\sqrt{2} > 1/2$: the central concentration is high relative to the overall spread. The density has a cusp, but the sign-test proof only requires the cdf to be differentiable at zero.

Example 2.6 (Logistic errors)**.** For logistic errors with cdf $F(x) = (1+e^{-x/s})^{-1}$ and scale $s > 0$, formula (2.3) gives:

$f(0)$	σ^2	$\mathrm{ARE}(\text{sign}, \text{mean})$
$1/(4s)$	$\pi^2 s^2/3$	$\pi^2/12 \approx .82247$

Exercise 6 verifies the variance formula.

The examples turn the slopes from Section 1 into concrete sample-size comparisons: a larger κ means fewer observations for the same local testing objective. They also raise the next question: could another test use the data more efficiently than either the mean or the sign test? Section 3 examines the likelihood and introduces Fisher information I. Section 4 then obtains a power bound for all tests; in the scalar regular problem, its one-sided power curve has slope $\sqrt{I}$. Section 5 connects this bound to Wald, Rao, and likelihood-ratio procedures.

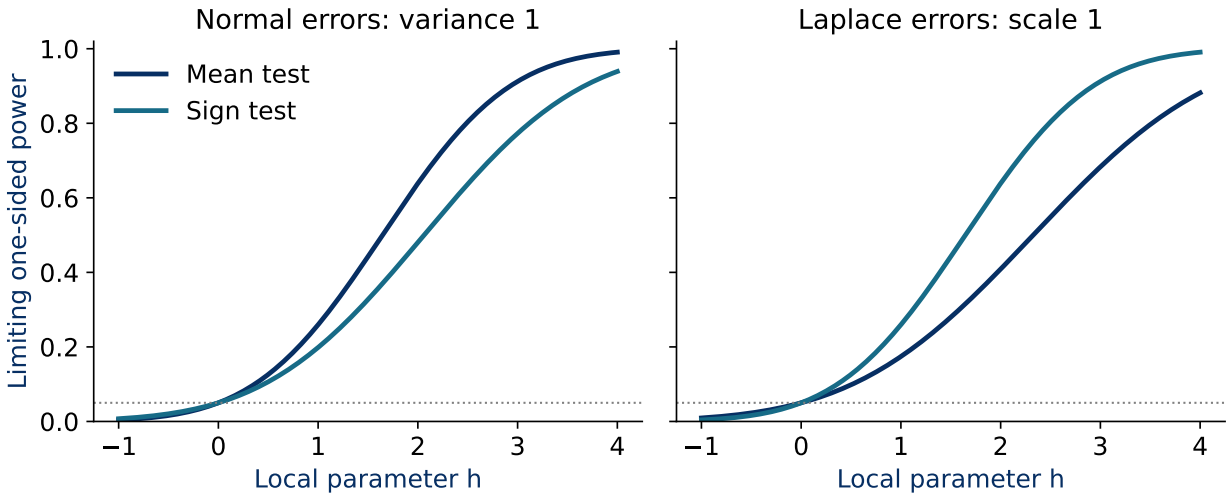


Figure 2: Local power at level .05 in two symmetric location families. The better local slope depends on the error distribution.

3 Local asymptotic normality

The sample-size comparison reduces efficiency to the local slope κ: a larger slope means fewer observations for the same power. For the mean and sign tests, we derived this slope using the translation structure of a location model. We now turn to the likelihood to derive local distributions in more general parametric models and ask how large the slope can be. A regular local likelihood has the form of a Gaussian likelihood, with information determining its curvature.

3.1 Likelihoods and the local experiment

A model is dominated by a measure μ if every P_θ has a density p_θ with respect to that same measure, so $P_\theta(A) = \int_A p_\theta \, d\mu$. Define the per-observation log likelihood, score, and Fisher information by

$$\ell_\theta = \log p_\theta, \qquad s_\theta = \nabla \ell_\theta, \qquad I(\theta) = \mathbb{E}_\theta[s_\theta s_\theta^T].$$

The gradient ∇ differentiates with respect to θ. The score is the random slope of the log likelihood; when its mean is zero, $I(\theta)$ is its covariance matrix. The total log likelihood is $\ell_n(\theta) = \sum_{i=1}^n \ell_\theta(X_i) = nP_n\ell_\theta$. We abbreviate $s_0 = s_{\theta_0}$ and $I = I(\theta_0)$. Information is per observation; total information is nI. When the score is centered and the regular likelihood information identity holds, taking $\psi_\theta = s_\theta$ in Section 0 gives $A = -I$, $B = I$, and influence function $I^{-1}s_0$, with covariance I^{-1}.

For probability measures P and Q on the same measurable space, write $\mathsf{Q} \ll \mathsf{P}$ when Q is *absolutely continuous with respect to* P: for every measurable set A, $\mathsf{P}(A) = 0$ implies $\mathsf{Q}(A) = 0$. By the Radon–Nikodym theorem, there is a nonnegative function $L = d\mathsf{Q}/d\mathsf{P}$, determined P-almost surely, such that

$$\mathsf{Q}(A) = \int_A L \, d\mathsf{P} \quad \text{for every measurable } A, \qquad \mathbb{E}_\mathsf{P} L = 1.$$

This function is the Radon–Nikodym derivative, or likelihood ratio. When both measures have densities p and q relative to the same measure, $L = q/p$ wherever $p > 0$.

Assume nearby distributions have a common positive support. Their likelihood ratio is

$$L_{n,h} = \frac{dP_{n,h}}{dP_{n,0}} = \exp\{\ell_n(\theta_n(h)) - \ell_n(\theta_0)\}, \qquad \mathbb{E}_{n,h} g(T_n) = \mathbb{E}_{n,0}[g(T_n) L_{n,h}]. \tag{3.1}$$

The expectation identity holds for every bounded measurable g and follows directly from the Radon–Nikodym definition. It explains why the local likelihood can determine the distribution of any statistic under a local alternative. Before using it asymptotically, we must control the likelihood ratio's tails.

Definition 3.1 (Statistical experiment)**.** A statistical experiment consists of a measurable sample space and an indexed family of probability measures on it. The local experiment here is $\mathcal{E}_n = \{P_{n,h} : h \in H_n\}$, where $H_n = \{h : \theta_0 + h/\sqrt{n} \in \Theta\}$. Every fixed $h \in \mathbb{R}^d$ belongs to H_n eventually. The candidate limit experiment is $\mathcal{G} = \{Q_h = N_d(h, I^{-1}) : h \in \mathbb{R}^d\}$. Here $I > 0$ means that I is symmetric positive definite: $u^T I u > 0$ for every nonzero u. The family $\mathcal{G}$ is a Gaussian shift experiment because h shifts the mean while the covariance stays fixed.

The Gaussian likelihood. The density of Q_h is proportional to $\exp\{-(y-h)^T I(y-h)/2\}$. Its normalizing constant is independent of h. Subtracting the exponent at zero gives

$$\begin{aligned} \log \frac{dQ_h}{dQ_0}(y) &= -\tfrac{1}{2}(y-h)^T I(y-h) + \tfrac{1}{2} y^T I y \\ &= h^T I y - \tfrac{1}{2} h^T I h. \end{aligned} \tag{3.2}$$

Under Q_0, $IY \sim N_d(0, I)$. Thus a random linear term with covariance I, followed by a deterministic quadratic term, characterizes this Gaussian likelihood.

Definition 3.2 (Root-n LAN and central sequence)**.** The model is locally asymptotically normal at θ_0 with positive definite information I if there are measurable random vectors $\Delta_n \in \mathbb{R}^d$, not depending on h, such that, for every fixed h,

$$\log L_{n,h} = h^T \Delta_n - \tfrac{1}{2} h^T I h + o_{\mathbb{P}}(1), \qquad \Delta_n \rightsquigarrow N_d(0, I) \tag{3.3}$$

under $P_{n,0}$. A sequence (Δ_n) satisfying these properties is called a *central sequence at* θ_0.

Remark 3.3 (Intuition). LAN approximates the local likelihood by the Gaussian likelihood in (3.2). The original observations need not be normal. The central sequence carries a noisy signal about h, and I determines the information available to distinguish that signal from zero.

The central sequence as accumulated score. For the regular iid models in Theorem 3.4, the central sequence is the normalized total score,

$$\Delta_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} s_0(X_i) = \frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0).$$

Each score records how one observation changes the log likelihood as the parameter moves from θ_0. Summing accumulates this evidence. Division by $\sqrt{n}$ keeps the random fluctuations on a fixed scale: centered scores with finite covariance I give $\Delta_n \rightsquigarrow N_d(0, I)$ by the CLT. The central sequence supplies the random linear term, while I supplies the deterministic curvature.

Example: a normal mean. If $X_i \sim N(\theta, 1)$, then $s_0(X_i) = X_i - \theta_0$, $I = 1$, and

$$\Delta_n = \sqrt{n}(\overline{X}_n - \theta_0), \qquad \log L_{n,h} = h\Delta_n - \tfrac{1}{2}h^2.$$

Indeed, expanding the normal log densities gives $\log L_{n,h} = (h/\sqrt{n})\sum_i (X_i - \theta_0) - h^2/2$. Under $P_{n,0}$, $\Delta_n \sim N(0,1)$; under $P_{n,h}$, $\Delta_n \sim N(h,1)$. This is the Gaussian local power calculation from Section 1, now read directly from the likelihood.

Under local alternatives, Theorem 3.8 will give $\Delta_n \rightsquigarrow N_d(Ih, I)$. In the scalar case, $\Delta_n/\sqrt{I}$ therefore has the slope $\sqrt{I}$ anticipated in Section 2. Sections 4–5 use this Gaussian shift to bound power and connect the central sequence to Rao, Wald, and likelihood-ratio tests.

3.2 Sufficient conditions and a uniform expansion

The definition gives an approximation at each fixed h. Later, likelihood-ratio testing evaluates the likelihood at a random estimated h. We therefore prove a version uniform on bounded local sets. The Hessian $\nabla^2 \ell_\theta$ is the matrix of second derivatives; the third derivative is a trilinear form whose norm is the supremum of its absolute value over three unit vectors. An envelope $M(X)$ below is one random upper bound valid for all parameters in the stated neighborhood.

Theorem 3.4 (Smooth iid LAN)**.** *Suppose an open ball $\mathcal{U}$ centered at θ_0 satisfies:*

1. *The densities p_θ have a common positive support, and $\ell_\theta(x)$ has three continuous derivatives in θ for almost every x in that support.*
2. *For $s_0 = \nabla \ell_{\theta_0}$ and $H_0 = \nabla^2 \ell_{\theta_0}$,*

$$\mathbb{E}_0 s_0 = 0, \quad \mathbb{E}_0 \|s_0\|^2 < \infty, \quad I = \mathbb{E}_0 s_0 s_0^T = -\mathbb{E}_0 H_0 > 0,$$

   and every entry of H_0 is integrable.
3. *A nonnegative M satisfies $\mathbb{E}_0 M(X) < \infty$ and $\sup_{\theta \in \mathcal{U}} \|\nabla^3 \ell_\theta(X)\| \le M(X)$ almost surely.*

Then LAN holds with $\Delta_n = n^{-1/2}\sum_i s_0(X_i)$. For each finite K,

$$\sup_{\|h\| \le K} \left| \ell_n(\theta_0 + h/\sqrt{n}) - \ell_n(\theta_0) - h^T\Delta_n + \tfrac{1}{2}h^T I h \right| \xrightarrow{\mathbb{P}} 0 \tag{3.4}$$

under $P_{n,0}$.

The common support makes change of measure exact. The information identity specifies the mean curvature; it follows by differentiating $\int p_\theta \, d\mu = 1$ twice when those differentiations can pass under the integral. We assume it explicitly rather than infer it from unverified domination. The envelope in condition 3 bounds the accumulation of third-order Taylor errors. Appendix C gives a weaker route to pointwise LAN through quadratic mean differentiability.

Proof of Theorem 3.4. Fix $K < \infty$. For all sufficiently large n, every segment from θ_0 to $\theta_0 + h/\sqrt{n}$,

$\|h\| \le K$, lies in $\mathcal{U}$. Taylor's theorem for each observation gives

$$\ell_n(\theta_0 + h/\sqrt{n}) - \ell_n(\theta_0) = h^T \Delta_n + \tfrac{1}{2} h^T \overline{H}_n h + R_n(h),$$
$$\overline{H}_n = \frac{1}{n} \sum_i H_0(X_i),$$
$$\sup_{\|h\| \le K} |R_n(h)| \le \sum_i \frac{K^3}{6n^{3/2}} M(X_i) = \frac{K^3}{6\sqrt{n}} P_n M.$$

The entrywise LLN and finite dimension imply $\|\overline{H}_n + I\| \xrightarrow{\mathbb{P}} 0$. The LLN also gives $P_n M \xrightarrow{\mathbb{P}} \mathbb{E}_0 M < \infty$. Hence the supremum error in (3.4) is bounded by

$$\tfrac{1}{2} K^2 \|\overline{H}_n + I\| + \frac{K^3}{6\sqrt{n}} P_n M \xrightarrow{\mathbb{P}} 0.$$

For any fixed $a \in \mathbb{R}^d$, the variables $a^T s_0(X_i)$ are iid, centered, and have variance $a^T I a$. The multivariate CLT gives $\Delta_n \rightsquigarrow N_d(0, I)$. Taking any fixed h in a bounded ball proves the LAN definition. □

Example 3.5 (Bernoulli LAN). For $X_i \sim \text{Bernoulli}(p)$, $p_0 \in (0,1)$,

$$s_0(x) = \frac{x - p_0}{p_0(1-p_0)}, \quad I = \frac{1}{p_0(1-p_0)}, \quad \Delta_n = \frac{\sqrt{n}(\overline{X}_n - p_0)}{p_0(1-p_0)}.$$

Indeed, $\ell_p(x) = x \log p + (1-x)\log(1-p)$, and $\text{Var}_{p_0}(X) = p_0(1-p_0)$ gives the displayed information. The support $\{0,1\}$ is common. On a sufficiently small interval around p_0 whose closure lies in $(0,1)$, all three derivatives are bounded uniformly in x. A constant envelope is integrable, so Theorem 3.4 applies.

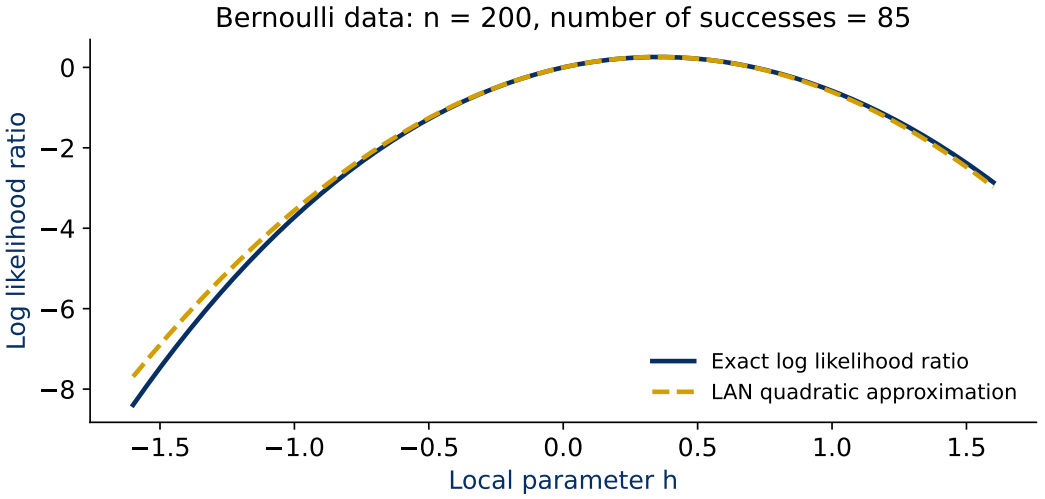


Figure 3: Exact Bernoulli local log likelihood ratio and its quadratic approximation, with $p_0 = .4$, $n = 200$, and 85 observed successes.

3.3 Contiguity and likelihood tilting

Likelihood tilting means reweighting the null law by the likelihood ratio to obtain an alternative law, as in (3.1). Under LAN the ratio converges in distribution, but it is unbounded. Passing the expectation identity to the limit therefore requires control of its large values. The mean of the likelihood limit provides that control and also yields contiguity, which will transfer estimator remainders to local alternatives.

Definition 3.6 (Contiguity)**.** Let P_n and Q_n be probability measures on the same measurable space at each n. The sequence Q_n is contiguous with respect to P_n, denoted $\mathsf{Q}_n \triangleleft \mathsf{P}_n$, if $\mathsf{P}_n(A_n) \to 0$ implies $\mathsf{Q}_n(A_n) \to 0$ for every sequence of measurable events A_n.

For instance, an error $r_n = o_{\mathbb{P}}(1)$ under P_n remains $o_{\mathbb{P}}(1)$ under Q_n: apply the definition to $A_n = \{\|r_n\| > \epsilon\}$ for each fixed $\epsilon > 0$. Absolute continuity concerns events of probability zero at each fixed n; contiguity concerns sequences of events whose probabilities tend to zero. The next lemma explains how a likelihood-ratio limit with mean one gives the latter property and also transfers limiting distributions.

Uniform integrability controls expectations contributed by very large values, uniformly over the index n. For the nonnegative likelihood ratios below, its precise tail condition is assertion 1. In the proof, $x \wedge K = \min\{x, K\}$ truncates from above, and $(x-K)_+ = \max\{x-K, 0\}$ is the excess above K.

Lemma 3.7 (Likelihood tilting and contiguity)**.** *Let $\mathsf{Q}_n \ll \mathsf{P}_n$ with $L_n = d\mathsf{Q}_n/d\mathsf{P}_n$. Suppose under P_n, $(T_n, L_n) \rightsquigarrow (T, L)$, where T_n, T are Euclidean-valued, $L \geq 0$, and $\mathbb{E}L = 1$. Then:*

1. *(L_n) is uniformly integrable: $\lim_{K\to\infty} \sup_n \mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > K\}] = 0$;*
2. *for bounded continuous g, $\mathbb{E}_{\mathsf{Q}_n} g(T_n) \to \mathbb{E}[g(T)L]$;*
3. *$\mathsf{Q}_n \triangleleft \mathsf{P}_n$.*

Proof of Lemma 3.7. Step 1: Uniform integrability. Because each L_n is a probability density, $\mathbb{E}_{\mathsf{P}_n} L_n = 1$. For fixed $K > 0$, the function $x \mapsto \min\{\max\{x, 0\}, K\}$ is bounded and continuous. Since $L_n \rightsquigarrow L$ and $L_n, L \geq 0$, the definition of weak convergence gives

$$\mathbb{E}_{\mathsf{P}_n}(L_n - K)_+ = 1 - \mathbb{E}_{\mathsf{P}_n}(L_n \wedge K) \longrightarrow 1 - \mathbb{E}(L \wedge K) = \mathbb{E}(L-K)_+.$$

Since $x\mathbf{1}\{x > 2K\} \leq 2(x-K)_+$ for $x \geq 0$, for every fixed $K > 0$,

$$\limsup_{n\to\infty} \mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > 2K\}] \leq 2\mathbb{E}(L-K)_+.$$

Since $L \geq 0$ and $\mathbb{E}L = 1 < \infty$, $L < \infty$ almost surely, so $(L-K)_+ \to 0$ almost surely as $K \to \infty$. Also, $0 \leq (L-K)_+ \leq L$. Dominated convergence therefore gives $\mathbb{E}(L-K)_+ \to 0$ as $K \to \infty$.

To include all indices n, fix $\epsilon > 0$. Choose $K_0 > 0$ with $2\mathbb{E}(L-K_0)_+ < \epsilon/4$. The preceding limsup bound gives some $N \geq 2$ such that $\mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > 2K_0\}] < \epsilon/2$ for every $n \geq N$. For each fixed $n < N$, dominated convergence applied to $L_n \mathbf{1}\{L_n > M\} \leq L_n$ gives $\mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > M\}] \to 0$ as $M \to \infty$. There are only finitely many such indices, so we can choose one $M_0 \geq 2K_0$ controls them all, and thus

$$\max\left\{ \max_{1\leq n<N} \mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > M_0\}],\ \sup_{n\geq N} \mathbb{E}_{\mathsf{P}_n}[L_n \mathbf{1}\{L_n > 2K_0\}] \right\} < \epsilon/2.$$

By monotonicity of the tail indicators, for every $M \geq M_0$ this maximum bounds all tail expectations. Their supremum over n is therefore less than $\epsilon/2$. Since $\epsilon > 0$ was arbitrary, assertion 1 follows.

Step 2: Convergence under the tilted law. Let $B = \|g\|_\infty = \sup_t |g(t)|$. Joint weak convergence applies to the bounded continuous function $(t, l) \mapsto g(t) \min\{\max\{l, 0\}, K\}$. Since $L_n, L \geq 0$, it gives, for every fixed K,

$$\mathbb{E}_{\mathsf{P}_n}[g(T_n)(L_n \wedge K)] \to \mathbb{E}[g(T)(L \wedge K)].$$

By change of measure and $L_n - (L_n \wedge K) = (L_n - K)_+$, the triangle inequality gives

$$\begin{aligned}|\mathbb{E}_{\mathsf{Q}_n} g(T_n) - \mathbb{E}[g(T)L]| \leq |\mathbb{E}_{\mathsf{P}_n}[g(T_n)(L_n \wedge K)] - \mathbb{E}[g(T)(L \wedge K)]| \\ + B\mathbb{E}_{\mathsf{P}_n}(L_n - K)_+ + B\mathbb{E}(L - K)_+.\end{aligned}$$

For fixed K, the truncated difference vanishes as $n \to \infty$, and the excess expectations converge by step 1. Hence

$$\limsup_{n\to\infty} |\mathbb{E}_{\mathsf{Q}_n} g(T_n) - \mathbb{E}[g(T)L]| \leq 2B\mathbb{E}(L - K)_+.$$

Now let $K \to \infty$. The dominated-convergence calculation in step 1 makes the right side tend to zero, proving assertion 2. The limit is expectation under the probability law $A \mapsto \mathbb{E}[L\mathbf{1}\{T \in A\}]$ on Borel sets, whose total mass is $\mathbb{E}L = 1$.

Step 3: Contiguity. For any events A_n and fixed K,

$$\mathsf{Q}_n(A_n) \leq K\mathsf{P}_n(A_n) + \mathbb{E}_{\mathsf{P}_n}[L_n\mathbf{1}\{L_n > K\}].$$

If $\mathsf{P}_n(A_n) \to 0$, its first term vanishes for fixed K; uniform integrability makes the remaining limsup arbitrarily small as $K \to \infty$. This proves assertion 3. □

The condition $\mathbb{E}L = 1$ prevents likelihood mass from escaping to rare events. The equality $\mathbb{E}L_n = 1$ alone would not justify interchanging limits and unbounded expectations.

Theorem 3.8 (Gaussian shift under local alternatives)**.** *Under LAN, for every fixed h, $P_{n,h} \triangleleft P_{n,0}$ and*

$$\Delta_n \rightsquigarrow N_d(Ih, I) \quad \text{under } P_{n,h}. \tag{3.5}$$

Consequently $I^{-1}\Delta_n \rightsquigarrow N_d(h, I^{-1})$ under that local alternative.

Proof of Theorem 3.8. Fix h. Under $P_{n,0}$, LAN gives

$$\log L_{n,h} = h^T\Delta_n - \tfrac{1}{2}h^T Ih + r_{n,h}, \qquad \Delta_n \rightsquigarrow Z \sim N_d(0, I), \qquad r_{n,h} \xrightarrow{\mathbb{P}} 0.$$

Since the remainder converges in probability to the constant zero, Slutsky's theorem (Lemma G.1, assertion 3) gives

$$(\Delta_n, r_{n,h}) \rightsquigarrow (Z, 0).$$

The map $(x, r) \mapsto (x, \exp\{h^T x - \frac{1}{2}h^T Ih + r\})$ is continuous. Applying the continuous mapping theorem to this joint limit yields

$$(\Delta_n, L_{n,h}) \rightsquigarrow (Z, L_h), \quad Z \sim N_d(0, I), \quad L_h = \exp(h^T Z - \tfrac{1}{2}h^T Ih).$$

The Gaussian moment-generating function yields $\mathbb{E}e^{h^T Z} = e^{h^T Ih/2}$, so $\mathbb{E}L_h = 1$. Lemma 3.7 therefore supplies contiguity and says that the local limit of Δ_n has expectations $\mathbb{E}[g(Z)L_h]$ for any continuous bounded function g.

Let f_0 denote the density of $Z \sim N_d(0, I)$:

$$f_0(z) = (2\pi)^{-d/2}(\det I)^{-1/2} \exp\{-\tfrac{1}{2} z^T I^{-1} z\},$$

where $\det I$ is the determinant of I. Define the weighted density

$$f_h(z) = f_0(z) \exp\{h^T z - \tfrac{1}{2} h^T I h\}.$$

This is the density that represents the limiting expectations, because

$$\mathbb{E}[g(Z)L_h] = \int_{\mathbb{R}^d} g(z) f_h(z)\, dz.$$

It is nonnegative and integrates to $\mathbb{E}L_h = 1$. To identify its distribution, combine the two exponential factors and complete the square:

$$-\tfrac{1}{2} z^T I^{-1} z + h^T z - \tfrac{1}{2} h^T I h = -\tfrac{1}{2}(z - Ih)^T I^{-1}(z - Ih).$$

Together with the factor $(2\pi)^{-d/2}(\det I)^{-1/2}$ from f_0, this shows that f_h is exactly the density of $N_d(Ih, I)$, proving (3.5). Multiplication by I^{-1} changes the mean to h and the covariance to $I^{-1} I I^{-1} = I^{-1}$. □

The next result applies to an estimator whose first-order term is $I^{-1}\Delta_n$. LAN is a condition on the model's likelihood; the estimator expansion below is a separate condition and does not by itself establish LAN. For smooth iid models, Theorem 3.4 establishes LAN under its stated conditions. The estimator's null asymptotic covariance is I^{-1}, the efficient covariance benchmark. Appendix F derives the scalar variance bound under its stated regularity conditions. Contiguity now transfers the estimator expansion to local alternatives, supplying the centered limit assumed in Corollary 1.9.

Corollary 3.9 (Efficient estimators under local alternatives)**.** *Assume the model is LAN at θ_0 with positive definite information I and central sequence Δ_n. Suppose under $P_{n,0}$ that $\sqrt{n}(\widehat{\theta}_n - \theta_0) = I^{-1}\Delta_n + o_{\mathbb{P}}(1)$. Then for every fixed h,*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \rightsquigarrow N_d(h, I^{-1}), \qquad \sqrt{n}(\widehat{\theta}_n - \theta_n(h)) \rightsquigarrow N_d(0, I^{-1})$$

under $P_{n,h}$. An information or covariance estimate consistent under $P_{n,0}$ remains consistent for its value at θ_0 under each $P_{n,h}$.

Proof of Corollary 3.9. Fix h and define the expansion error by

$$r_n = \sqrt{n}(\widehat{\theta}_n - \theta_0) - I^{-1}\Delta_n,$$

and thus $r_n \xrightarrow{\mathbb{P}} 0$ under $P_{n,0}$ by the assumed expansion. For each fixed $\epsilon > 0$, define $A_n = \{\|r_n\| > \epsilon\}$. Then $P_{n,0}(A_n) \to 0$. By Theorem 3.8, LAN gives $P_{n,h} \triangleleft P_{n,0}$. Applying the definition of contiguity (Definition 3.6) to these events yields

$$P_{n,h}(\|r_n\| > \epsilon) = P_{n,h}(A_n) \longrightarrow 0.$$

Since this holds for every $\epsilon > 0$, $r_n = o_{\mathbb{P}}(1)$ also under $P_{n,h}$. The error r_n is the same function of the sample under both laws; only the law used to compute its probability has changed. Thus the identity

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = I^{-1}\Delta_n + r_n$$

also has a vanishing remainder under the local alternative. Theorem 3.8 gives $I^{-1}\Delta_n \rightsquigarrow N_d(h, I^{-1})$ under $P_{n,h}$. Adding $r_n = o_{\mathbb{P}}(1)$ preserves that limit by Lemma G.1. Since $\sqrt{n}(\theta_n(h) - \theta_0) = h$, subtracting h proves the second limit. For an estimate $\widehat{C}_n \xrightarrow{\mathbb{P}} C$ under $P_{n,0}$, apply Definition 3.6 to the events $\{\|\widehat{C}_n - C\| > \epsilon\}$ to get $P_{n,h}(\|\widehat{C}_n - C\| > \epsilon) \to 0$ for every $\epsilon > 0$. Hence $\widehat{C}_n \xrightarrow{\mathbb{P}} C$ also under $P_{n,h}$. □

The likelihood argument has now verified the local centering required by Corollary 1.9. In a scalar model, an efficient standardized estimator, or the standardized central sequence, has local slope $\sqrt{I}$. To decide whether any test can improve on it, we must compare arbitrary sample decisions with decisions in the Gaussian experiment. This is the role of the representation theorem.

4 Asymptotic optimality of tests

The central sequence has a Gaussian local limit, and its standardized scalar form has slope $\sqrt{I}$. A competing test can nevertheless use the entire sample. A power bound must account for that freedom. The representation theorem does so by showing that every subsequential local power function is the power function of a randomized test in the Gaussian experiment. We can then solve one Gaussian testing problem to bound all original tests.

4.1 The asymptotic representation theorem

The limiting Gaussian experiment consists of observing one random vector $Y \in \mathbb{R}^d$. Under local parameter h, its law is $Q_h = N_d(h, I^{-1})$. This is the local limiting law of the standardized score $I^{-1}\Delta_n$ under $P_{n,h}$, by Theorem 3.8. The theorem below constructs a test of Y that reproduces the limiting powers of an arbitrary original test sequence.

We use tightness in its Euclidean-space sense: a sequence Y_n is tight if, for every $\epsilon > 0$, some compact set K satisfies $\sup_n \mathbb{P}(Y_n \notin K) < \epsilon$. Lemma G.1 supplies boundedness in probability from a distributional limit; the proof below includes the finite-index step needed for tightness of the full family.

Theorem 4.1 (LAN representation of tests)**.** *Assume LAN as in Definition 3.2. For any measurable tests $0 \le \phi_n \le 1$ and any subsequence, there are a further subsequence and a measurable $\phi_\infty : \mathbb{R}^d \to [0,1]$ such that, along that further subsequence,*

$$\mathbb{E}_{n,h}\phi_n \longrightarrow \mathbb{E}_{Q_h}\phi_\infty(Y), \qquad Y \sim Q_h = N_d(h, I^{-1}), \tag{4.1}$$

for every fixed $h \in \mathbb{R}^d$.

Remark 4.2 (Intuition). Theorem 4.1 represents the limiting power of a test based on the full sample by the power of one test based only on one Gaussian observation. Here is an explicit example of the two tests. Let X_i be iid $N(h/\sqrt{n}, 1)$, with $n \ge 2$, and write $Y_n = \sqrt{n}\bar{X}_n \sim N(h, 1)$, where $\bar{X}_n$ is the sample mean. For a fixed threshold c, consider

$$\phi_n = \mathbf{1}\{Y_n > c\}\mathbf{1}\{X_1 > X_2\}.$$

This test uses both the sample mean and the ordering of the first two observations. The difference $X_1 - X_2 \sim N(0, 2)$ is independent of Y_n: their covariance is zero, and they are jointly

normal. Thus $X_1 > X_2$ has probability $1/2$, independently of Y_n, under every h. In the Gaussian experiment, observe only $Y \sim Q_h = N(h, 1)$ and use the fixed test function

$$\phi_\infty(y) = \tfrac{1}{2}\mathbf{1}\{y > c\}.$$

It rejects with probability $1/2$ when $y > c$ and otherwise accepts. The two tests have the same power:

$$\mathbb{E}_{n,h}\phi_n = \tfrac{1}{2}[1 - \Phi(c - h)] = \mathbb{E}_{Q_h}\phi_\infty(Y).$$

This is exactly the representation in (4.1), with $I = 1$; here equality holds for every n, so the asserted convergence follows immediately. Theorem 4.1 gives such a power representation for arbitrary test sequences in LAN models, along a suitable subsequence.

The following compactness result supplies the joint limit used in the proof.

Lemma 4.3 (Finite-dimensional Prokhorov theorem)**.** *Let W_n be random vectors in $\mathbb{R}^k$ for a fixed $k \geq 1$. If their laws are tight, then every subsequence has a further subsequence W_{n_j} such that $W_{n_j} \rightsquigarrow W$ for some $\mathbb{R}^k$-valued random vector W.*

Proof of Theorem 4.1. Extract one joint null limit. Under $P_{n,0}$, $\Delta_n \rightsquigarrow N_d(0, I)$, so Δ_n is tight. For each $\epsilon > 0$, choose a compact ball $K \subset \mathbb{R}^d$ with $\limsup_n P_{n,0}(\Delta_n \notin K) < \epsilon$. Since $\phi_n \in [0, 1]$, the pair lies in $K \times [0, 1]$ whenever $\Delta_n \in K$. Enlarging the compact set to cover the finitely many early indices gives tightness of the full sequence of pair laws. Applying Lemma 4.3 to $(\Delta_n, \phi_n) \in \mathbb{R}^{d+1}$ along any prescribed subsequence gives a further subsequence with

$$(\Delta_n, \phi_n) \rightsquigarrow (Z, U), \qquad Z \sim N_d(0, I), \quad 0 \leq U \leq 1.$$

The bound on U follows because $\mathbb{R}^d \times [0, 1]$ is closed and has probability one at every n.

Construct the limiting test. Let ν be the law of Z, and define a finite measure m on Borel sets by $m(B) = \mathbb{E}[U\mathbf{1}\{Z \in B\}]$. The inequalities $0 \leq U \leq 1$ give $0 \leq m(B) \leq \nu(B)$, so $\nu(B) = 0$ implies $m(B) = 0$. The Radon–Nikodym theorem, aka Theorem G.4 in Appendix G, therefore gives a density $r = dm/d\nu$ that we choose to take values in $[0, 1]$. For every Borel set $B \subset \mathbb{R}^d$, the defining integral identity gives

$$\mathbb{E}[r(Z)\mathbf{1}\{Z \in B\}] = \int_B r(z)\,\nu(dz) = m(B) = \mathbb{E}[U\mathbf{1}\{Z \in B\}].$$

Since $r(Z)$ is a measurable function of Z, this is the defining property of $r(Z) = \mathbb{E}[U \mid Z]$, the conditional average of the limiting rejection variable. Set

$$\phi_\infty(y) = r(Iy).$$

Transfer the same limit to a fixed alternative. Fix h. By LAN, write $L_{n,h} = \exp\{h^T\Delta_n - \frac{1}{2}h^T Ih + \rho_{n,h}\}$, where $\rho_{n,h} = o_{\mathbb{P}}(1)$ under $P_{n,0}$. Along the already selected subsequence, Lemma G.1 gives $(\Delta_n, \phi_n, \rho_{n,h}) \rightsquigarrow (Z, U, 0)$. Apply the continuous mapping theorem with

$$f_h(z, u, t) = \big(z, u, \exp\{h^T z - \tfrac{1}{2}h^T Ih + t\}\big).$$

This continuous function maps $(\Delta_n, \phi_n, \rho_{n,h})$ to $(\Delta_n, \phi_n, L_{n,h})$, so

$$(\Delta_n, \phi_n, L_{n,h}) \rightsquigarrow (Z, U, e^{h^T Z - h^T Ih/2}).$$

Since $Z \sim N_d(0, I)$, the scalar $h^T Z$ is normal with mean zero and variance $h^T I h$. Its moment-generating function gives

$$\mathbb{E}e^{h^T Z - h^T I h/2} = e^{-h^T I h/2}\mathbb{E}e^{h^T Z} = e^{-h^T I h/2} e^{h^T I h/2} = 1.$$

We can therefore apply Lemma 3.7 with $T_n = (\Delta_n, \phi_n)$ and the bounded continuous clipped-coordinate function $g(z, u) = \max\{0, \min\{1, u\}\}$. Its value is ϕ_n on the sample pairs and U on the limit. Thus

$$\begin{aligned}\mathbb{E}_{n,h}\phi_n &\longrightarrow \mathbb{E}[U e^{h^T Z - h^T I h/2}] \\ &= \mathbb{E}[r(Z) e^{h^T Z - h^T I h/2}] \\ &= \mathbb{E}_{Z_h \sim N_d(Ih, I)} r(Z_h) = \mathbb{E}_{Q_h}\phi_\infty(Y).\end{aligned}$$

The first equality is the tower property: the exponential factor is a function of Z, so

$$\mathbb{E}[U e^{h^T Z - h^T I h/2}] = \mathbb{E}[e^{h^T Z - h^T I h/2}\mathbb{E}[U \mid Z]] = \mathbb{E}[r(Z) e^{h^T Z - h^T I h/2}].$$

The second equality is the Gaussian tilt calculated in Theorem 3.8. For the third equality, set $Y = I^{-1} Z_h$. Then $Y \sim N_d(h, I^{-1}) = Q_h$ and $r(Z_h) = r(IY) = \phi_\infty(Y)$ by the definition of ϕ_∞.

The extraction depended only on (Δ_n, ϕ_n) under the null, before h was fixed. For each fixed h, LAN applies to that same subsequence; no further extraction depending on h occurred. This proves the common-subsequence assertion. Finally, if the full power functions have pointwise limits, every subsequence has those same limits, so the Gaussian test just constructed represents that limiting function. □

Remark 4.4 (Scope of the representation). The theorem makes a subsequential statement about rejection probabilities. It assumes neither an estimator expansion nor convergence of the full original decision sequence. Its simultaneous validity for each fixed h does not assert convergence uniformly over h.

4.2 Gaussian optimality and the power envelope

For a scalar model, write $Q_h = N(h, I^{-1})$ with $I > 0$. The Gaussian null is $h \le 0$, and the alternatives are $h > 0$.

Definition 4.5 (Uniformly most powerful test)**.** A test is *uniformly most powerful* (UMP) at level α if it has level α and its power is at least that of every other level-α test at every alternative. The word uniformly means that the same test is optimal for all alternatives.

The resulting upper bound on power, as a function of h, is called the power envelope.

Proposition 4.6 (Gaussian UMP test)**.** *For $Y \sim Q_h = N(h, I^{-1})$, the test $\phi^*_\infty(Y) = \mathbf{1}\{\sqrt{I}\, Y > z_{1-\alpha}\}$ has level α over $h \le 0$ and is UMP among tests of that level. Its power is*

$$\beta^*(h) = 1 - \Phi(z_{1-\alpha} - h\sqrt{I}). \tag{4.2}$$

In fact, for every $h > 0$, it dominates all tests whose rejection probability at $h = 0$ is at most α.

Proof of Proposition 4.6. Under Q_h, $\sqrt{I}\,Y \sim N(h\sqrt{I}, 1)$, giving (4.2). This power increases in h and equals α at zero. Hence

$$\sup_{t\le 0} \mathbb{E}_{Q_t}\phi^*_\infty = \mathbb{E}_{Q_0}\phi^*_\infty = \alpha,$$

which establishes level α over the composite null.

Fix $h > 0$ and put $a = z_{1-\alpha}/\sqrt{I}$. The likelihood ratio $L_h(y) = (dQ_h/dQ_0)(y) = \exp(hIy - h^2 I/2)$ is strictly increasing. Let $c = L_h(a) > 0$, so $\phi^*_\infty(y) = \mathbf{1}\{L_h(y) > c\}$ for every y. For any $0 \le \phi_\infty \le 1$, the following pointwise inequality always holds:

$$(\phi_\infty - \phi^*_\infty)(L_h - c) \le 0$$

Indeed, above the threshold its first factor is nonpositive and its second is positive; below the threshold their signs are reversed; at the threshold the second factor is zero. If $\mathbb{E}_{Q_0}\phi_\infty \le \alpha$, change of measure and the decomposition $L_h = (L_h - c) + c$ give

$$\begin{aligned}\mathbb{E}_{Q_h}\phi_\infty - \mathbb{E}_{Q_h}\phi^*_\infty &= \mathbb{E}_{Q_0}[(\phi_\infty - \phi^*_\infty)L_h] \\ &= \mathbb{E}_{Q_0}[(\phi_\infty - \phi^*_\infty)(L_h - c)] + c\{\mathbb{E}_{Q_0}\phi_\infty - \alpha\} \le 0.\end{aligned}$$

Here we used $\mathbb{E}_{Q_0}\phi^*_\infty = \alpha$. The first term in the last line is nonpositive by the pointwise inequality; the second is nonpositive because $c > 0$ and $\mathbb{E}_{Q_0}\phi_\infty \le \alpha$. All expectations are finite since the tests are bounded and $\mathbb{E}_{Q_0}L_h = 1$. This is the Neyman–Pearson inequality, proved here for the needed Gaussian problem. Every competing test of level α over $t \le 0$ satisfies $\mathbb{E}_{Q_0}\phi_\infty \le \alpha$, so this power bound applies to it. The candidate ϕ^*_∞ has the required composite-null level, as shown above. Although $c = L_h(a)$ depends on h, its rejection region $\{y > a\}$ does not. Thus the same test dominates every competitor at every $h > 0$ and is UMP.

□

4.3 Local asymptotically uniformly most powerful tests

The representation now turns the Gaussian calculation into a bound on every original test, whether or not its full power function converges.

Theorem 4.7 (Scalar local asymptotic power envelope)**.** *Assume scalar LAN at θ_0 with $I > 0$. If $0 \le \phi_n \le 1$ and $\limsup_n \mathbb{E}_{n,0}\phi_n \le \alpha$, then for every fixed $h > 0$,*

$$\limsup_n \mathbb{E}_{n,h}\phi_n \le \beta^*(h). \tag{4.3}$$

Proof of Theorem 4.7. Fix $h > 0$ and let $b = \limsup_{n\to\infty} \mathbb{E}_{n,h}\phi_n$. Since the powers lie in $[0, 1]$, there is a strictly increasing sequence (n_k) such that

$$\mathbb{E}_{n_k,h}\phi_{n_k} \longrightarrow b \qquad \text{as } k \to \infty.$$

Apply Theorem 4.1 to this subsequence. It gives a further subsequence (n_{k_j}) and a measurable test $\phi_\infty : \mathbb{R} \to [0, 1]$ such that

$$\lim_{j\to\infty} \mathbb{E}_{n_{k_j},t}\phi_{n_{k_j}} = \mathbb{E}_{Q_t}\phi_\infty \qquad \text{for every fixed } t \in \mathbb{R}.$$

In particular, at $t = 0$,

$$\mathbb{E}_{Q_0}\phi_\infty = \lim_{j\to\infty} \mathbb{E}_{n_{k_j},0}\phi_{n_{k_j}} \le \limsup_{n\to\infty} \mathbb{E}_{n,0}\phi_n \le \alpha.$$

The limit is along the extracted subsequence, whereas the limsup is over the full original sequence. At $t = h$, taking a further subsequence preserves the already established limit b, so

$$b = \lim_{j\to\infty} \mathbb{E}_{n_{k_j},h}\phi_{n_{k_j}} = \mathbb{E}_{Q_h}\phi_\infty \le \beta^*(h).$$

The final inequality is the last assertion of Proposition 4.6, which requires only $\mathbb{E}_{Q_0}\phi_\infty \le \alpha$. Since $h > 0$ was arbitrary, this proves (4.3). The subsequences and the represented test may depend on the initially chosen h; the bound holds separately at each fixed positive alternative. □

Definition 4.8 (Local AUMP)**.** A sequence of measurable tests $0 \le \phi_n \le 1$ has local asymptotic level α for $h \le 0$ if $\limsup_n \mathbb{E}_{n,h}\phi_n \le \alpha$ for every fixed $h \le 0$. A sequence ϕ_n^* is locally asymptotically uniformly most powerful if it has that property, its power converges at every fixed $h \in \mathbb{R}$, and for every competing sequence ϕ_n of that local level and every fixed $h > 0$,

$$\lim_n \mathbb{E}_{n,h}\phi_n^* \ge \limsup_n \mathbb{E}_{n,h}\phi_n.$$

The same sequence must be optimal at every fixed positive local alternative; uniform convergence in h is not required by this definition.

Corollary 4.9 (Locally AUMP score test)**.** *Under scalar LAN at θ_0 with $I > 0$, the tests $\phi_n^* = \mathbf{1}\{\Delta_n/\sqrt{I} > z_{1-\alpha}\}$ are locally AUMP in the sense of Definition 4.8.*

Proof of Corollary 4.9. Fix $t \in \mathbb{R}$. Theorem 3.8 gives $\Delta_n/\sqrt{I} \rightsquigarrow Z_t \sim N(t\sqrt{I}, 1)$ under $P_{n,t}$. Since Z_t has a density, $\mathbb{P}(Z_t = z_{1-\alpha}) = 0$. Lemma 1.6(ii), applied to the upper-tail set $(z_{1-\alpha}, \infty)$, therefore gives

$$\lim_{n\to\infty} \mathbb{E}_{n,t}\phi_n^* = \mathbb{P}(Z_t > z_{1-\alpha}) = 1 - \Phi(z_{1-\alpha} - t\sqrt{I}) = \beta^*(t).$$

Thus the candidate's power converges at every fixed t. For $t \le 0$, $\beta^*(t) \le \beta^*(0) = \alpha$, proving its local asymptotic level.

For any competing sequence (ϕ_n) of that local level, $\limsup_n \mathbb{E}_{n,0}\phi_n \le \alpha$. Theorem 4.7 consequently gives, for every fixed $h > 0$,

$$\limsup_{n\to\infty} \mathbb{E}_{n,h}\phi_n \le \beta^*(h) = \lim_{n\to\infty} \mathbb{E}_{n,h}\phi_n^*.$$

These are exactly the conditions in Definition 4.8. □

This proves optimality in the local problem. Uniform size over an unrestricted global null $\theta \le \theta_0$ requires a model-specific argument, such as the monotonicity argument for a location statistic in Section 1.4.

4.4 Two-sided alternatives

Now consider the point null $h = 0$ against $h \ne 0$. The optimal Gaussian region for any fixed $h > 0$ is the upper tail; for any fixed $h < 0$ it is the lower tail. These are unique up to Q_0-null sets. To see uniqueness in the preceding inequality, equality of power forces both nonpositive terms to vanish. Away from $L_h = c$, the pointwise product can then vanish only if $\phi_\infty = \phi_\infty^*$. The boundary has zero probability. The two tail regions differ on a set of positive Gaussian probability for $0 < \alpha < 1$, so no test can be UMP for every $h \ne 0$ without additional restrictions on the competitor class.

For independent $Z_1, \dots, Z_r \sim N(0,1)$ and fixed $m \in \mathbb{R}^r$, the law of $\sum_{j=1}^r (Z_j + m_j)^2$ is the noncentral chi-squared law $\chi_r^2(\lambda)$, with degrees of freedom r and noncentrality $\lambda = \|m\|^2$. At $\lambda = 0$ it is the central law χ_r^2. Write $q_{r,u}$ for its lower-tail quantile, defined by $\mathbb{P}\{\chi_r^2 \le q_{r,u}\} = u$ for $0 < u < 1$.

The two-sided Gaussian rule uses $IY^2 > q_{1,1-\alpha}$. Its power is

$$\mathbb{P}\{\chi_1^2(h^2 I) > q_{1,1-\alpha}\},$$

because the square of $N(h\sqrt{I}, 1)$ is $\chi_1^2(h^2 I)$. It addresses departures in both directions. The next section derives this same quadratic statistic from score, estimator displacement, and maximized likelihood gain, and separates its two-sided use from the one-sided AUMP result.

5 Wald, Rao, and likelihood ratio tests

The power envelope (4.2) identifies the slope $\sqrt{I}$. Corollary 4.9 gives a score test that attains it. In applications, the same local information can be read from an estimated parameter or from a maximized likelihood. We now derive those connections. Rao's statistic uses the likelihood slope at the null, Wald's statistic uses the fitted displacement, and the likelihood ratio (LR) uses the fitted gain. Here first-order equivalence means that the statistics differ by $o_{\mathbb{P}}(1)$ under the specified laws. It follows from one local quadratic likelihood, rather than from three unrelated normal approximations.

5.1 Scalar score and Wald statistics

For a scalar point null $\theta = \theta_0$, define

$$U_n(\theta) = \sum_i s_\theta(X_i), \qquad Z_{S,n} = \frac{U_n(\theta_0)}{\sqrt{nI}} = \frac{\Delta_n}{\sqrt{I}}, \qquad S_n = Z_{S,n}^2.$$

Rao's score statistic uses only the null fit. If an estimator has the efficient expansion and $\widehat{I}_n \xrightarrow{\mathbb{P}} I > 0$ under $P_{n,0}$, Wald's signed and quadratic statistics are

$$Z_{W,n} = \sqrt{n\widehat{I}_n}(\widehat{\theta}_n - \theta_0), \qquad W_n = Z_{W,n}^2.$$

For example, $\widehat{I}_n = I(\widehat{\theta}_n)$ is consistent if $I(\cdot)$ is continuous and $\widehat{\theta}_n$ is consistent. Assign information-based statistics a fixed value on exceptional events where a required positive square root or inverse does not exist. Consistency makes these events negligible.

Proposition 5.1 (Score and Wald local limits)**.** *Assume scalar LAN with score central sequence $\Delta_n = n^{-1/2} U_n(\theta_0)$, the efficient expansion $\sqrt{n}(\widehat{\theta}_n - \theta_0) = I^{-1}\Delta_n + o_{\mathbb{P}}(1)$ under $P_{n,0}$, and $\widehat{I}_n \xrightarrow{\mathbb{P}} I > 0$ under $P_{n,0}$. For every fixed h, under $P_{n,h}$,*

$$Z_{S,n} \rightsquigarrow N(h\sqrt{I}, 1), \qquad Z_{W,n} - Z_{S,n} = o_{\mathbb{P}}(1), \qquad W_n - S_n = o_{\mathbb{P}}(1).$$

In particular, $W_n, S_n \rightsquigarrow \chi_1^2(h^2 I)$.

Proof of Proposition 5.1. Theorem 3.8 gives the local normal limit of $Z_{S,n}$. Work first under the null. Let $r_n = \sqrt{n}(\widehat{\theta}_n - \theta_0) - I^{-1}\Delta_n = o_{\mathbb{P}}(1)$, so $\sqrt{n}(\widehat{\theta}_n - \theta_0) = \Delta_n/I + r_n$. On the event $\widehat{I}_n > 0$, substituting this identity into the definitions gives

$$\begin{aligned} Z_{W,n} - Z_{S,n} &= \sqrt{\widehat{I}_n}\,\sqrt{n}(\widehat{\theta}_n - \theta_0) - \frac{\Delta_n}{\sqrt{I}} \\ &= \sqrt{\widehat{I}_n}\left(\frac{\Delta_n}{I} + r_n\right) - \frac{\Delta_n}{\sqrt{I}} \\ &= \left(\frac{\sqrt{\widehat{I}_n}}{I} - \frac{1}{\sqrt{I}}\right)\Delta_n + \sqrt{\widehat{I}_n}\, r_n \\ &= \frac{\sqrt{\widehat{I}_n} - \sqrt{I}}{I}\Delta_n + \sqrt{\widehat{I}_n}\, r_n. \end{aligned}$$

Information consistency and the continuous mapping theorem give $\sqrt{\widehat{I}_n} - \sqrt{I} = o_{\mathbb{P}}(1)$ and $\sqrt{\widehat{I}_n} = O_{\mathbb{P}}(1)$. Also, LAN gives $\Delta_n \rightsquigarrow N(0, I)$ under the null, hence $\Delta_n = O_{\mathbb{P}}(1)$. Thus the two terms in the last line are respectively $o_{\mathbb{P}}(1)O_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)o_{\mathbb{P}}(1)$, both $o_{\mathbb{P}}(1)$ by Lemma G.1. Since $P_{n,0}(\widehat{I}_n \leq 0) \to 0$, the fixed definition on the exceptional event does not change the conclusion $Z_{W,n} - Z_{S,n} = o_{\mathbb{P}}(1)$. Now $Z_{W,n} + Z_{S,n} = 2Z_{S,n} + o_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$, giving

$$W_n - S_n = (Z_{W,n} - Z_{S,n})(Z_{W,n} + Z_{S,n}) = o_{\mathbb{P}}(1).$$

Fix h and write $e_n = Z_{W,n} - Z_{S,n}$. We have proved that $P_{n,0}(|e_n| > \epsilon) \to 0$ for every $\epsilon > 0$. Theorem 3.8 gives $P_{n,h} \triangleleft P_{n,0}$, so the definition of contiguity implies

$$P_{n,h}(|e_n| > \epsilon) \longrightarrow 0 \qquad \text{for every } \epsilon > 0.$$

Thus $e_n = o_{\mathbb{P}}(1)$ also under $P_{n,h}$. The same argument transfers $W_n - S_n = o_{\mathbb{P}}(1)$ and information consistency to that local alternative.

Under $P_{n,h}$, Theorem 3.8 gives $Z_{S,n} \rightsquigarrow N(h\sqrt{I}, 1)$. Since $e_n \xrightarrow{\mathbb{P}} 0$ under the same laws, Slutsky's theorem (Lemma G.1) yields

$$Z_{W,n} = Z_{S,n} + e_n \rightsquigarrow N(h\sqrt{I}, 1) \qquad \text{under } P_{n,h}.$$

Finally, if $Z \sim N(m, 1)$, then Z^2 has the noncentral chi-squared law $\chi^2_1(m^2)$. Applying the continuous mapping theorem to the square of each signed statistic therefore gives $S_n, W_n \rightsquigarrow \chi^2_1(h^2 I)$ under $P_{n,h}$, as claimed. □

For one-sided testing use a signed statistic; squaring removes the direction. Replacing I in the score denominator by a consistent null information estimate changes the signed statistic by $o_{\mathbb{P}}(1)$, by exactly the product calculation above.

5.2 Deriving the MLE expansion under the smooth assumptions

To apply the preceding proposition to a likelihood fit, we now derive its assumed expansion. The following lemma is stated for dimension d because the same calculation will also treat nuisance parameters.

Lemma 5.2 (Consistent interior MLE expansion)**.** *Assume that all conditions of Theorem 3.4 hold in dimension d. Let $\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ under $P_{n,0}$ and suppose that, with $P_{n,0}$-probability tending to one, it is an interior score root: $U_n(\widehat{\theta}_n) = \nabla \ell_n(\widehat{\theta}_n) = 0$. Then*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = I^{-1}\Delta_n + o_{\mathbb{P}}(1).$$

A maximization assumption is not needed for this expansion; consistency and the score equation suffice.

Proof of Lemma 5.2. Let $\delta_n = \widehat{\theta}_n - \theta_0$. With probability tending to one, the score equation holds and the entire segment $\theta_0 + t\delta_n$, $0 \le t \le 1$, lies in $\mathcal{U}$. This follows from consistency and the fact that $\mathcal{U}$ is an open ball around θ_0. On that event, the fundamental theorem of calculus gives the exact vector identity

$$0 = \frac{1}{n}U_n(\theta_0) + \mathcal{H}_n\delta_n, \qquad \mathcal{H}_n = \int_0^1 \frac{1}{n}\sum_i \nabla^2 \ell_{\theta_0 + t\delta_n}(X_i)\, dt.$$

The third-derivative envelope implies the Hessian Lipschitz bound

$$\|\nabla^2 \ell_{\theta_0+t\delta_n}(X_i) - H_0(X_i)\| \le t\|\delta_n\| M(X_i).$$

After averaging and integrating,

$$\|\mathcal{H}_n - P_n H_0\| \le \tfrac{1}{2}\|\delta_n\| P_n M = o_{\mathbb{P}}(1).$$

The last equality uses consistency and $P_n M = O_{\mathbb{P}}(1)$. Since $P_n H_0 \xrightarrow{\mathbb{P}} -I$, we have $\mathcal{H}_n \xrightarrow{\mathbb{P}} -I$. On the vanishing-probability exceptional event, define $\mathcal{H}_n = -I$; this does not alter that convergence. Lemma G.2 gives invertibility with probability tending to one and $\mathcal{H}_n^{-1} \xrightarrow{\mathbb{P}} -I^{-1}$. Solving the exact score identity on the good event yields

$$\sqrt{n}\delta_n = -\mathcal{H}_n^{-1}\Delta_n = I^{-1}\Delta_n + o_{\mathbb{P}}(1),$$

because $\Delta_n = O_{\mathbb{P}}(1)$. This establishes the root-n rate as a consequence of the argument; it was not assumed to control the Hessian error. □

5.3 Likelihood ratio and its local limit

The MLE expansion locates the likelihood maximizer. Evaluating the uniform LAN expansion there will give its gain in log likelihood. Return to the scalar point null and define

$$\Lambda_n = 2\{\ell_n(\widehat{\theta}_n) - \ell_n(\theta_0)\},$$

where $\widehat{\theta}_n$ is an unrestricted maximizer. Consistency of this maximizer is a separate assumption, not a consequence of the local expansion alone.

Theorem 5.3 (Regular scalar LR and first-order equivalence)**.** *Assume that all conditions of Theorem 3.4 hold in dimension one. Suppose an unrestricted maximizer $\widehat{\theta}_n$ is consistent under $P_{n,0}$ and is an interior score root with $P_{n,0}$-probability tending to one. Then under the null and every fixed local alternative,*

$$\Lambda_n = \Delta_n^2/I + o_{\mathbb{P}}(1).$$

If $\widehat{I}_n \xrightarrow{\mathbb{P}} I$ under $P_{n,0}$, the LR, score, and Wald quadratic statistics are pairwise equal up to $o_{\mathbb{P}}(1)$ and have common limit $\chi^2_1(h^2 I)$ under $P_{n,h}$.

Proof of Theorem 5.3. By Lemma 5.2, $\widehat{h}_n = \sqrt{n}(\widehat{\theta}_n - \theta_0) = \Delta_n/I + o_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$ under the null. By Theorem 3.4, specifically (3.4), the LAN expansion is uniform on bounded local sets. Write its remainder as

$$R_n(t) = \ell_n(\theta_0 + t/\sqrt{n}) - \ell_n(\theta_0) - t\Delta_n + \tfrac{1}{2}It^2.$$

For every finite K, $\sup_{|t|\le K} |R_n(t)| \xrightarrow{\mathbb{P}} 0$. To justify the random substitution, for $\epsilon > 0$,

$$P_{n,0}(|R_n(\widehat{h}_n)| > \epsilon) \le P_{n,0}(|\widehat{h}_n| > K) + P_{n,0}\left(\sup_{|t|\le K} |R_n(t)| > \epsilon\right).$$

Tightness makes the limsup of the first term arbitrarily small by choosing K large, and the second tends to zero for that fixed K. Hence $R_n(\widehat{h}_n) = o_{\mathbb{P}}(1)$. Pointwise LAN without the bounded-set uniformity would not supply this inequality's second limit.

We can therefore complete the local square:

$$\begin{aligned}\ell_n(\widehat{\theta}_n) - \ell_n(\theta_0) &= \widehat{h}_n\Delta_n - \tfrac{1}{2}I\widehat{h}_n^2 + o_{\mathbb{P}}(1) \\ &= \frac{\Delta_n^2}{2I} - \frac{I}{2}(\widehat{h}_n - \Delta_n/I)^2 + o_{\mathbb{P}}(1) = \frac{\Delta_n^2}{2I} + o_{\mathbb{P}}(1).\end{aligned}$$

The squared difference is $o_{\mathbb{P}}(1)$ because $\widehat{h}_n - \Delta_n/I = o_{\mathbb{P}}(1)$. Multiply by two. Contiguity transfers the resulting remainder to every $P_{n,h}$. Since $S_n = \Delta_n^2/I$ exactly, and Proposition 5.1 gives $W_n - S_n = o_{\mathbb{P}}(1)$ and their local limit, all the conclusions follow. □

Remark 5.4 (Intuition). The same local parabola links three measurements: its slope at the null, the displacement of its maximum, and the height gained at that maximum. The proofs identify the errors in those links and show that each error vanishes under the null and local alternatives.

For $q_{r,u}$ as defined in Section 4, the common scalar two-sided local power is

$$\mathbb{P}\{\chi^2_1(h^2 I) > q_{1,1-\alpha}\}. \tag{5.1}$$

A noncentral chi-squared law here has no atom at the positive critical value, so Lemma 1.6(ii) justifies the rejection-probability limit. Formula (5.1) equals (1.3) with $\kappa = \sqrt{I}$, since $q_{1,1-\alpha} = z^2_{1-\alpha/2}$. Signed score and Wald tests attain the one-sided envelope. The unrestricted quadratic LR statistic instead addresses the two-sided problem considered here.

5.4 Bernoulli likelihood tests

The Bernoulli model from Example 3.5 provides a complete calculation of the three statistics. Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and test $p = p_0 \in (0,1)$ against $p \ne p_0$. Put $\widehat{p} = \overline{X}_n$. The per-observation information is $I(p) = 1/[p(1-p)]$.

For an interior observed estimate, $0 < \widehat{p} < 1$, Wald's statistic is

$$W_n = \frac{n(\widehat{p} - p_0)^2}{\widehat{p}(1-\widehat{p})}. \tag{5.2}$$

The total score at p_0 is $n(\widehat{p} - p_0)/[p_0(1-p_0)]$. Dividing its square by total null information gives

$$S_n = \frac{n(\widehat{p} - p_0)^2}{p_0(1-p_0)}. \tag{5.3}$$

For the LR, write $\ell_n(p) = n\{\widehat{p}\log p + (1-\widehat{p})\log(1-p)\}$. Its derivative vanishes at $p = \widehat{p}$, and its second derivative is negative for an interior sample proportion. Subtracting $\ell_n(p_0)$ from $\ell_n(\widehat{p})$ gives

$$\Lambda_n = 2n\left[\widehat{p}\log\frac{\widehat{p}}{p_0} + (1-\widehat{p})\log\frac{1-\widehat{p}}{1-p_0}\right]. \tag{5.4}$$

For the LR formula, use the continuous convention $0\log 0 = 0$ at sample boundaries. The ordinary plug-in Wald formula is not defined there. For $p_n = p_0 + h/\sqrt{n}$, choose $\epsilon > 0$ so eventually $p_n \in [\epsilon, 1-\epsilon]$. The boundary probability is $p_n^n + (1-p_n)^n \le 2(1-\epsilon)^n \to 0$, which justifies treating these events as negligible in the local theorem.

Example 3.5 verifies the smooth LAN conditions. The Bernoulli sample mean is consistent and is an interior score root outside the negligible boundary events above. Proposition 5.1 and Theorem 5.3 therefore give the common limit under $p = p_0 + h/\sqrt{n}$:

$$\chi^2_1\left(\frac{h^2}{p_0(1-p_0)}\right).$$

With $n = 100$, 52 successes, and $p_0 = .40$, the two-sided calculations are below. Each approximate p-value is the null χ^2_1 upper-tail probability beyond the observed statistic.

Test	Statistic	χ^2_1 upper-tail p-value
Wald	5.769231	0.016309
Rao score	6.000000	0.014306
LR	5.864103	0.015453

All three exceed $q_{1,.95} = 3.841459$. These are approximate p-values. First-order equivalence does not make the finite-sample formulas identical.

For the upper-tail problem $p \le p_0$ against $p > p_0$, use the signed score

$$Z_{S,n} = \frac{\sqrt{n}(\widehat{p} - p_0)}{\sqrt{p_0(1-p_0)}}.$$

The row CLT gives $Z_{S,n} \rightsquigarrow N(h/\sqrt{p_0(1-p_0)}, 1)$, so Theorem 1.7 gives local power $1 - \Phi(z_{1-\alpha} - h/\sqrt{p_0(1-p_0)})$. Construct $K_n(p) = \sum_i \mathbf{1}\{U_i \le p\}$ for all p using the same iid uniform variables U_i. This common construction, called a coupling, makes the count nondecreasing in p for every realization. Thus an upper-tail count test has its largest null rejection probability at p_0. Exact binomial calibration can provide exact global level if desired.

The efficiency comparison revisited. For an asymptotically linear estimator in the scalar LAN model, local regularity means that its centered limiting law is unchanged by any fixed local shift h. Under this condition, Proposition F.1 proves that its asymptotic variance satisfies $V \ge I^{-1}$. A consistently studentized test based on this estimator therefore has slope $1/\sqrt{V} \le \sqrt{I}$. Under the calibrated-sequence assumptions of Theorem 2.2, its ARE relative to an efficient test is $1/(VI) \le 1$. The local power calculation began with an unspecified slope κ. The likelihood analysis has now identified its efficient value $\sqrt{I}$, and the Gaussian bound shows why that value cannot be improved in the stated local problem. Estimator variance, Pitman sample size, and one-sided power therefore express the same information constraint under their respective regularity conditions.

Exercises and solutions

These exercises connect the local power calculations to sample-size planning and likelihood geometry.

Exercise 1 (local slope)**.** A statistic has local limit $N(1.5h, 1)$. At one-sided level .05, find its limiting power at $h = 1$, its ARE relative to a slope-one test, and its planned sample size for shift .2 and target power .80.

Solution to Exercise 1. The power is $1 - \Phi(1.644854 - 1.5) \approx .4424$. The ARE is $1.5^2 = 2.25$, so its sample-size multiplier relative to the slope-one test is $1/2.25$. Formula (2.1) gives $n \approx 68.70$, rounded up to 69. □

Exercise 2 (direction and two-sided power)**.** A statistic has local limit $N(h, 1)$. Find the two-sided local power at level .05, and compare its limiting power for h and $-h$.

Solution to Exercise 2. Formula (1.3) gives $1 - \Phi(1.959964 - h) + \Phi(-1.959964 - h)$. Symmetry of the standard normal distribution makes this an even function of h. The upper-tail one-sided power does not have this symmetry because that rule targets positive alternatives. □

Exercise 3 (Exact Gaussian LAN)**.** Let $X_i \overset{\text{iid}}{\sim} N(\theta, 4)$ and test $\theta = 0$ against $\theta > 0$. Find s_0, I, the central sequence, the exact local likelihood ratio, and the optimal local power at $h = 2$ and level .05.

Solution to Exercise 3. The log density is $\ell_\theta(x) = -\log(8\pi)/2 - (x - \theta)^2/8$, so $s_0(x) = x/4$ and $I = \mathbb{E}_0(X/4)^2 = 4/16 = 1/4$. Thus $\Delta_n = \sqrt{n}\overline{X}_n/4$. Expanding the squares gives

$$\ell_n(h/\sqrt{n}) - \ell_n(0) = -\frac{1}{8}\sum_i \{(X_i - h/\sqrt{n})^2 - X_i^2\} = \frac{h}{4\sqrt{n}}\sum_i X_i - \frac{h^2}{8} = h\Delta_n - \frac{h^2}{8}.$$

There is no remainder. The standardized score is $\Delta_n/\sqrt{I} = \sqrt{n}\overline{X}_n/2$. Under the null it is exactly $N(0, 1)$, so reject above $z_{.95} = 1.644854$. Under $\theta_n = h/\sqrt{n}$ it is exactly $N(h/2, 1)$. At $h = 2$ the rejection probability is $1 - \Phi(1.644854 - 1) \approx .2595$, attaining the envelope $1 - \Phi(z_{.95} - h\sqrt{I})$. □

Supplementary proofs and reading

Overview. The supplements give exact discrete calibration, a median representation, weaker conditions for LAN, and extensions to vector targets and nuisance parameters. They also supply shared probability tools and justify the minimum-sample-size interpretation of local efficiency.
Proof guide. Appendix A supports the finite-sample sign example. The median supplement connects sign testing to influence functions. Quadratic mean differentiability (DQM) supplies a weaker route to LAN in Appendix C. Appendix D extends the scalar likelihood tests; its final example treats an unknown variance. Appendix E gives Gaussian optimality with nuisance directions. The influence-function proof identifies the variance constraint behind the testing envelope. Appendix G supplies the small-remainder, matrix-inversion, and Bernoulli limit arguments. Appendix H proves uniform local power for the location tests and then derives the limits of their minimum sample sizes.

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A Exact calibration of the sign test

Section 1.4 uses a normal cutoff for asymptotic level. Here we choose a binomial cutoff and boundary rejection probability to obtain exact level at each sample size. This also supplies the calibrated sign test used in Appendix H. Write $\mathbb{P}_p$ for probabilities under $K_n \sim \mathrm{Bin}(n, p)$, where K_n is the number of positive observations. Choose an integer $k_n \in \{0, \dots, n\}$ such that

$$\mathbb{P}_{1/2}(K_n > k_n) \le \alpha \le \mathbb{P}_{1/2}(K_n \ge k_n), \qquad \rho_n = \frac{\alpha - \mathbb{P}_{1/2}(K_n > k_n)}{\mathbb{P}_{1/2}(K_n = k_n)}.$$

Such an integer exists because consecutive binomial upper tails bracket every $\alpha \in (0, 1)$, and the mass at each possible count is positive.

Proposition A.1 (Exact level and local power of the calibrated sign test)**.** *Under the sign-model assumptions of Proposition 1.12, $0 \le \rho_n \le 1$, and*

$$\phi_n = \mathbf{1}\{K_n > k_n\} + \rho_n \mathbf{1}\{K_n = k_n\} \tag{A.1}$$

has exact level α for $\theta \le 0$. Its local power is the same as that of the normal-cutoff sign test.

Proof of Proposition A.1. Exact calibration and composite-null level. The cutoff inequalities make the numerator in ρ_n lie between zero and its denominator. Substitution in (A.1) gives $\mathbb{E}_{\theta=0}\phi_n = \alpha$. For $\theta \le 0$, symmetry gives success probability $p = F(\theta) \le 1/2$. Using independent uniforms

U_i, construct the counts $K_n(p) = \sum_i \mathbf{1}\{U_i \le p\}$. Then $K_n(p)$ is nondecreasing in p for every realization. The conditional rejection function of the count takes the values $0, \rho_n, 1$ in increasing order, so its expectation is nondecreasing in p. It is therefore at most its boundary value α for every null parameter, and the maximum equals α at zero.

Convergence of the integer cutoff. Put $t_n = 2(k_n - n/2)/\sqrt{n}$, $d_n = 2/\sqrt{n}$, and $z = z_{1-\alpha}$. Under the null let $Z_n = 2(K_n - n/2)/\sqrt{n} \rightsquigarrow N(0,1)$ by Lemma G.3. The calibration inequalities are

$$\mathbb{P}_{1/2}(Z_n > t_n) \le \alpha \le \mathbb{P}_{1/2}(Z_n > t_n - d_n).$$

Fix $\epsilon > 0$. If $t_n \le z - \epsilon$ along infinitely many indices, the left tail probability is at least $\mathbb{P}_{1/2}(Z_n > z - \epsilon) \to 1 - \Phi(z - \epsilon) > \alpha$, contradicting the first inequality. If $t_n \ge z + \epsilon$ along infinitely many indices, then eventually $t_n - d_n \ge z + \epsilon/2$. The probability on the right is then at most $\mathbb{P}_{1/2}(Z_n > z + \epsilon/2) \to 1 - \Phi(z + \epsilon/2) < \alpha$, contradicting the second inequality. Thus $t_n \to z$.

Vanishing boundary contribution. Under each fixed $P_{n,h}$, Proposition 1.12 gives $T_n^{\text{sign}} \rightsquigarrow Z_h \sim N(2f(0)h, 1)$. Since $t_n \to z$, Lemma G.1 gives $T_n^{\text{sign}} - t_n \rightsquigarrow Z_h - z$. Apply Lemma 1.6(ii) with $X_n = T_n^{\text{sign}} - t_n$, $X = Z_h - z$, and $D = (0, \infty)$; the limiting variable has zero probability at $\partial D = \{0\}$. Thus

$$P_{n,h}(K_n > k_n) = P_{n,h}(T_n^{\text{sign}} > t_n) \to \mathbb{P}(Z_h > z).$$

For fixed $\epsilon > 0$, eventually $K_n = k_n$ implies $|T_n^{\text{sign}} - z| \le \epsilon$. The closed-set bound in Lemma 1.6(i), with $X_n = T_n^{\text{sign}}$, $X = Z_h$, and $F = [z - \epsilon, z + \epsilon]$, gives

$$\limsup_n P_{n,h}(K_n = k_n) \le \mathbb{P}(|Z_h - z| \le \epsilon).$$

Let $\epsilon \downarrow 0$; the right side vanishes because Z_h has no atom at z. Finally, $0 \le \rho_n P_{n,h}(K_n = k_n) \le P_{n,h}(K_n = k_n) \to 0$. Adding the two parts of (A.1) proves the local power claim. □

For $n = 100$ and $\alpha = .05$, binomial tail summation gives $k_n = 58$ and $\rho_n = 0.2551091$. Under normal errors of variance one and $\theta = .25$, $p = \Phi(.25)$, so

$$\beta_{\text{sign}} = \sum_{k=59}^{100} \binom{100}{k} p^k (1-p)^{100-k} + \rho_n \binom{100}{58} p^{58}(1-p)^{42} \approx 0.631545.$$

For comparison, the known-variance mean power is 0.803765. The sign local approximation is 0.636777. Studentizing the mean preserves its local limit and gives a different finite-sample power function.

B The median connection

The sign statistic counts crossings of zero. The median moves until the empirical count reaches one half. This gives the two procedures the same first-order local sensitivity. The proof obtains the required local empirical-cdf bound using a finite grid, monotonicity, and Chebyshev's inequality.

Proposition B.1 (Median limit and linear representation)**.** *Let ε_i be iid with continuous cdf F, unique median zero, and derivative $f(0) > 0$ at zero. Write $\varepsilon_{(k)}$ for the kth smallest observation. Let*

$\widehat{m}_n = \varepsilon_{(k_n)}$ be an order-statistic median with $k_n - n/2 = O(1)$. Then

$$\sqrt{n}\widehat{m}_n \rightsquigarrow N\left(0, \frac{1}{4f(0)^2}\right),$$

and

$$\sqrt{n}\widehat{m}_n = \frac{1}{f(0)\sqrt{n}} \sum_i \left[\tfrac{1}{2} - \mathbf{1}\{\varepsilon_i \le 0\}\right] + o_{\mathbb{P}}(1). \tag{B.1}$$

For any real sequence θ_n and $X_i = \theta_n + \varepsilon_i$, the corresponding median has the same centered representation with $\widehat{m}_n - \theta_n$ on the left.

Proof of Proposition B.1. Distributional limit and rate. For fixed real t, the event $\sqrt{n}\widehat{m}_n \le t$ means that at least k_n observations are at most $t/\sqrt{n}$. Its count is binomial with parameter $q_n = F(t/\sqrt{n}) = 1/2 + f(0)t/\sqrt{n} + o(n^{-1/2})$. By Lemma G.3, its centered, standardized value Z_n converges to $Z_0 \sim N(0,1)$. The count event is $Z_n > a_n$, where

$$a_n = \frac{k_n - 1 - nq_n}{\sqrt{nq_n(1-q_n)}} \longrightarrow -2f(0)t.$$

Lemma G.1 gives $Z_n - a_n \rightsquigarrow Z_0 + 2f(0)t$. Apply Lemma 1.6(ii) with $X_n = Z_n - a_n$, $X = Z_0 + 2f(0)t$, and $D = (0,\infty)$. Its boundary $\{0\}$ has zero probability under the limiting normal law, so the count event has limiting probability $1 - \Phi(-2f(0)t) = \Phi(2f(0)t)$. This proves the stated normal limit and hence $\widehat{m}_n = O_{\mathbb{P}}(n^{-1/2})$.

Control of the empirical cdf near zero. Let $F_n(u) = n^{-1}\sum_i \mathbf{1}\{\varepsilon_i \le u\}$ be the empirical cdf and define

$$H_n(u) = \sqrt{n}\{(F_n - F)(u) - (F_n - F)(0)\}.$$

We show that, for each fixed K, $\sup_{|u|\le K/\sqrt{n}} |H_n(u)| \xrightarrow{\mathbb{P}} 0$. Choose a fixed integer $J \ge 1$ and grid points $u_j = -K/\sqrt{n} + 2Kj/(J\sqrt{n})$, $0 \le j \le J$. At each point, $H_n(u_j)$ is centered and

$$\mathrm{Var}(H_n(u_j)) \le |F(u_j) - F(0)| = O(n^{-1/2}).$$

This variance bound follows by writing the increment as an average of signed indicators of the interval between zero and u_j. Chebyshev's inequality and a union bound over the finite grid imply $\max_j |H_n(u_j)| \xrightarrow{\mathbb{P}} 0$ for fixed J.

For $u_j \le u \le u_{j+1}$, monotonicity of F_n and F gives

$$H_n(u_j) - \sqrt{n}\{F(u_{j+1}) - F(u_j)\} \le H_n(u) \le H_n(u_{j+1}) + \sqrt{n}\{F(u_{j+1}) - F(u_j)\}.$$

Differentiability at zero means $F(u) = 1/2 + f(0)u + r(u)$ with $\sup_{|u|\le K/\sqrt{n}} \sqrt{n}|r(u)| \to 0$. Thus

$$\sup_{|u|\le K/\sqrt{n}} |H_n(u)| \le \max_j |H_n(u_j)| + \frac{2Kf(0)}{J} + o(1).$$

For any desired positive tolerance, first choose J large enough to control the deterministic term, then let $n \to \infty$. This proves the claimed local supremum bound.

Solve the empirical median equation. Tightness of $\sqrt{n}\widehat{m}_n$ lets us choose K so $\mathbb{P}(|\widehat{m}_n| > K/\sqrt{n})$ is arbitrarily small. Combining that localization with the preceding supremum bound gives $H_n(\widehat{m}_n) = o_{\mathbb{P}}(1)$. Continuity of F implies no ties almost surely, so $F_n(\widehat{m}_n) = k_n/n = 1/2 + O(1/n)$. Also,

$$\sqrt{n}\{F(\widehat{m}_n) - 1/2\} = f(0)\sqrt{n}\widehat{m}_n + o_{\mathbb{P}}(1),$$

because $\widehat{m}_n = O_{\mathbb{P}}(n^{-1/2})$ and the deterministic differentiability remainder is $o(|\widehat{m}_n|)$. Inserting these identities into

$$F_n(\widehat{m}_n) - 1/2 = F_n(0) - 1/2 + F(\widehat{m}_n) - F(0) + \{(F_n - F)(\widehat{m}_n) - (F_n - F)(0)\}$$

and multiplying by $\sqrt{n}$ yields $0 = \sqrt{n}(F_n(0) - 1/2) + f(0)\sqrt{n}\widehat{m}_n + o_{\mathbb{P}}(1)$. Rearrangement gives (B.1). Order statistics commute exactly with translations, proving the last assertion. □

The influence function in (B.1) has mean zero and variance $1/[4f(0)^2]$, since the indicator at zero is Bernoulli$(1/2)$. A consistently studentized median test therefore has slope $2f(0)$ by Corollary 1.9, matching the sign test.

C Quadratic mean differentiability and a full LAN proof

Theorem 3.4 obtains LAN by differentiating the log density three times. Quadratic mean differentiability (DQM) instead approximates its square root in mean square, so it can apply when the log density has a kink. For a measure ν, $L^2(\nu)$ is the space of functions with finite squared norm $\|g\|^2_{L^2(\nu)} = \int |g|^2 \, d\nu$; for a vector, sum the squared component norms. The result below proves fixed-h LAN. The stronger uniform expansion from the main text remains the route used to evaluate the likelihood at a random fitted parameter.

Definition C.1 (DQM at θ_0)**.** A dominated model is differentiable in quadratic mean at θ_0 if there is a measurable vector $s_0 \in L^2(P_{\theta_0})^d$ such that, as $t \to 0$,

$$\int \left[\sqrt{p_{\theta_0+t}} - \sqrt{p_{\theta_0}} - \tfrac{1}{2} t^T s_0 \sqrt{p_{\theta_0}}\right]^2 d\mu = o(\|t\|^2). \tag{C.1}$$

The norm in (C.1) is an $L^2(\mu)$ norm. The condition says that the map $\theta \mapsto \sqrt{p_\theta}$ has a linear approximation in that norm. When ordinary smooth score calculus is valid, the derivative is $s_0\sqrt{p_{\theta_0}}/2$.

Theorem C.2 (DQM implies iid LAN with common positive support)**.** *Assume DQM at the interior point θ_0, $\mathbb{E}_0 s_0 = 0$, and $I = \mathbb{E}_0 s_0 s_0^T > 0$. Suppose the nearby densities are positive on the same support. Then for every fixed h,*

$$\log \frac{dP^{\otimes n}_{\theta_0 + h/\sqrt{n}}}{dP^{\otimes n}_{\theta_0}} = h^T \Delta_n - \tfrac{1}{2} h^T I h + o_{\mathbb{P}}(1), \qquad \Delta_n = \frac{1}{\sqrt{n}} \sum_i s_0(X_i) \rightsquigarrow N_d(0, I).$$

Proof of Theorem C.2. Fix $h \neq 0$; the case $h = 0$ is immediate. Put $t_n = h/\sqrt{n}$ and, under $P_0 = P_{\theta_0}$, define

$$Y_n(x) = \sqrt{p_{\theta_0+t_n}(x)/p_{\theta_0}(x)} - 1, \quad a_n(x) = \frac{h^T s_0(x)}{2\sqrt{n}}, \quad R_n(x) = Y_n(x) - a_n(x).$$

Because the common positive support permits division by p_{θ_0}, multiplying the squared residual by p_{θ_0} and integrating turns the DQM norm into

$$n\mathbb{E}_0 R_n^2 = o(1). \tag{C.2}$$

Let $q = h^T I h$. Here $\mathbb{E}_0 a_n = 0$ and $n\mathbb{E}_0 a_n^2 = q/4$. The identity $Y_n = a_n + R_n$ and Cauchy–Schwarz give

$$\left| n\mathbb{E}_0 Y_n^2 - \frac{q}{4} \right| \le 2\{n\mathbb{E}_0 a_n^2\}^{1/2}\{n\mathbb{E}_0 R_n^2\}^{1/2} + n\mathbb{E}_0 R_n^2 \to 0.$$

Consequently

$$n\mathbb{E}_0 Y_n^2 \longrightarrow q/4. \tag{C.3}$$

Also, normalization of the density gives the exact identity

$$\mathbb{E}_0(1 + Y_n)^2 = 1, \quad \text{so} \quad \mathbb{E}_0 Y_n = -\tfrac{1}{2}\mathbb{E}_0 Y_n^2.$$

Because $\mathbb{E}_0 a_n = 0$, the same identity gives $\mathbb{E}_0 R_n = -\mathbb{E}_0 Y_n^2/2$.

The linear sum. By (C.2), $\mathrm{Var}_{n,0}(\sum_i R_n(X_i)) \le n\mathbb{E}_0 R_n^2 = o(1)$. Chebyshev's inequality therefore gives

$$\sum_i R_n(X_i) - n\mathbb{E}_0 R_n = o_{\mathbb{P}}(1), \qquad n\mathbb{E}_0 R_n = -\tfrac{n}{2}\mathbb{E}_0 Y_n^2 \to -q/8.$$

Since $\sum_i a_n(X_i) = h^T \Delta_n/2$, it follows that

$$\sum_i Y_n(X_i) = \tfrac{1}{2} h^T \Delta_n - \tfrac{1}{8} q + o_{\mathbb{P}}(1). \tag{C.4}$$

The quadratic sum. For every $\epsilon > 0$, Markov's inequality gives

$$P_{n,0}\left(\sum_i R_n(X_i)^2 > \epsilon \right) \le n\mathbb{E}_0 R_n^2/\epsilon \to 0.$$

Thus $\sum_i R_n(X_i)^2 = o_{\mathbb{P}}(1)$. The iid LLN gives

$$\sum_i a_n(X_i)^2 = \frac{1}{4n}\sum_i (h^T s_0(X_i))^2 \xrightarrow{\mathbb{P}} q/4.$$

The cross term is bounded by $2(\sum_i a_n^2)^{1/2}(\sum_i R_n^2)^{1/2} = o_{\mathbb{P}}(1)$. Hence

$$\sum_i Y_n(X_i)^2 \xrightarrow{\mathbb{P}} q/4. \tag{C.5}$$

The largest summand. The residual maximum is at most $(\sum_i R_n^2)^{1/2} = o_{\mathbb{P}}(1)$. For the linear part, put $W_i = h^T s_0(X_i)$, let W denote a variable with their common law, and note $\mathbb{E}_0 W^2 < \infty$. For each $\epsilon > 0$,

$$P_{n,0}\left(\max_{i \le n} |W_i| > \epsilon\sqrt{n} \right) \le n P_{\theta_0}(|W| > \epsilon\sqrt{n}) \le \epsilon^{-2}\mathbb{E}_0[W^2 \mathbf{1}\{|W| > \epsilon\sqrt{n}\}] \to 0.$$

Thus $\max_i |Y_n(X_i)| = o_{\mathbb{P}}(1)$.

The log likelihood. With probability tending to one, all $|Y_n(X_i)| \le 1/2$. On this event, a deterministic Taylor bound gives $|2\log(1+y) - 2y + y^2| \le (16/3)|y|^3$: the third derivative of $2\log(1+y)$

is $4/(1+y)^3$, whose absolute value is at most 32 on that interval. The sum of the remainders is at most $(16/3)\max_i |Y_n(X_i)| \sum_i Y_n(X_i)^2 = o_{\mathbb{P}}(1)$. Therefore

$$\begin{aligned}\log L_{n,h} &= 2\sum_i \log(1+Y_n(X_i)) \\ &= 2\sum_i Y_n(X_i) - \sum_i Y_n(X_i)^2 + o_{\mathbb{P}}(1) \\ &= h^T\Delta_n - \tfrac{1}{4}q - \tfrac{1}{4}q + o_{\mathbb{P}}(1) \\ &= h^T\Delta_n - \tfrac{1}{2}q + o_{\mathbb{P}}(1).\end{aligned}$$

The multivariate CLT gives the asserted central-sequence limit. □

The theorem proves pointwise LAN at each fixed local parameter. Evaluating the local likelihood expansion at a random maximizer needs further uniformity or likelihood arguments; the smooth assumptions in Theorem 5.3 already provide that control.

D Vector restrictions and nuisance parameters

Section 5 treats a scalar target with one standard error and one likelihood curvature. For several restrictions, these become a covariance matrix and a quadratic form. A nuisance parameter is an unknown component of the model that is not the target of the hypothesis. Adjusting for it removes score variation that can be reproduced by changing that component. We first extend Wald standardization to vectors, then derive this nuisance adjustment and compare all three likelihood tests.

D.1 Wald tests for vector restrictions

Let $g : \Theta \to \mathbb{R}^r$ describe the null $g(\theta) = 0$. Its derivative $Dg(\theta)$ is the $r \times d$ Jacobian matrix, with entries $\partial g_i/\partial\theta_j$. Row rank r means that the r rows are linearly independent, so the restrictions are locally independent. The covariance V below may be a likelihood covariance or a sandwich covariance.

For a symmetric positive definite matrix $C = Q\,\mathrm{diag}(\lambda_1,\dots,\lambda_r)Q^T$ with orthogonal Q and $\lambda_j > 0$, define $C^{\pm 1/2} = Q\,\mathrm{diag}(\lambda_1^{\pm 1/2},\dots,\lambda_r^{\pm 1/2})Q^T$. Transforming a vector with covariance C by $C^{-1/2}$ gives identity covariance; this standardization is called whitening.

Theorem D.1 (Wald test for smooth restrictions)**.** *Suppose g is continuously differentiable near θ_0, $g(\theta_0) = 0$, and $G = Dg(\theta_0)$ has row rank r. Under each fixed $P_{n,h}$ assume*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \rightsquigarrow N_d(h, V), \qquad \widehat{V}_n \xrightarrow{\mathbb{P}} V > 0.$$

Set $\widehat{G}_n = Dg(\widehat{\theta}_n)$ and

$$W_n = n g(\widehat{\theta}_n)^T [\widehat{G}_n \widehat{V}_n \widehat{G}_n^T]^{-1} g(\widehat{\theta}_n)$$

when the inverse exists, and assign a fixed value otherwise. Then

$$W_n \rightsquigarrow \chi^2_r(\lambda), \qquad \lambda = (Gh)^T(GVG^T)^{-1}Gh.$$

At $h = 0$, the limit is central χ^2_r.

Proof of Theorem D.1. Fix h. The assumed local limit implies $\sqrt{n}(\widehat{\theta}_n - \theta_0) = O_{\mathbb{P}}(1)$ and hence $\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$. Differentiability gives $g(\theta_0 + t) = Gt + r(t)$ with $\|r(t)\|/\|t\| \to 0$. Therefore, as justified in Lemma G.2,

$$Z_n := \sqrt{n} g(\widehat{\theta}_n) = G\sqrt{n}(\widehat{\theta}_n - \theta_0) + o_{\mathbb{P}}(1) \rightsquigarrow Z \sim N_r(Gh, C), \qquad C = GVG^T.$$

For nonzero $a \in \mathbb{R}^r$, full row rank gives $G^T a \neq 0$, and $a^T C a = (G^T a)^T V (G^T a) > 0$. Thus C is positive definite. Continuity of Dg yields $\widehat{G}_n \xrightarrow{\mathbb{P}} G$, so $C_n = \widehat{G}_n \widehat{V}_n \widehat{G}_n^T \xrightarrow{\mathbb{P}} C$. Lemma G.2 gives invertibility with probability tending to one and $C_n^{-1} \xrightarrow{\mathbb{P}} C^{-1}$.

Since Z_n is tight,

$$|Z_n^T C_n^{-1} Z_n - Z_n^T C^{-1} Z_n| \leq \|Z_n\|^2 \|C_n^{-1} - C^{-1}\| = o_{\mathbb{P}}(1).$$

Continuous mapping gives $Z_n^T C^{-1} Z_n \rightsquigarrow Z^T C^{-1} Z$. The vector $C^{-1/2} Z$ has mean $C^{-1/2} Gh$ and identity covariance. Its squared Euclidean norm is, by definition, $\chi^2_r(\|C^{-1/2} Gh\|^2)$. This is the claimed λ. The exceptional event does not affect the limit because its probability tends to zero. □

When an efficient estimator is used, Corollary 3.9 supplies $V = I^{-1}$. A correctly estimated sandwich covariance can instead calibrate a Wald test for a regular estimating-equation target. That covariance calculation by itself does not supply the likelihood score identity or the likelihood power envelope.

D.2 Nuisance parameters and Rao's efficient score

Partition $\theta = (\vartheta, \eta)$ with target $\vartheta \in \mathbb{R}^r$ and nuisance $\eta \in \mathbb{R}^k$, so $d = r + k$. Test $\vartheta = \vartheta_0$ and write the score and information at $\theta_0 = (\vartheta_0, \eta_0)$ as

$$s = \begin{pmatrix} s_\vartheta \\ s_\eta \end{pmatrix}, \qquad I = \begin{pmatrix} I_{\vartheta\vartheta} & I_{\vartheta\eta} \\ I_{\eta\vartheta} & I_{\eta\eta} \end{pmatrix} > 0.$$

The vectors s_ϑ and s_η are the target and nuisance blocks of the per-observation score. Write $U_\vartheta(\theta)$ and $U_\eta(\theta)$ for the corresponding blocks of the total score $U_n(\theta) = \sum_i s_\theta(X_i)$, suppressing n in the block notation. Partition the central sequence as $\Delta_n = (\Delta_{\vartheta n}^T, \Delta_{\eta n}^T)^T = n^{-1/2} U_n(\theta_0)$. Define the efficient target score and its information by

$$s_{\text{eff}} = s_\vartheta - I_{\vartheta\eta} I_{\eta\eta}^{-1} s_\eta, \quad J := I_{\vartheta \cdot \eta} = I_{\vartheta\vartheta} - I_{\vartheta\eta} I_{\eta\eta}^{-1} I_{\eta\vartheta}. \tag{D.1}$$

Here $I_{\eta\eta} > 0$ as a principal submatrix. For nonzero u, the vector $(u, -I_{\eta\eta}^{-1} I_{\eta\vartheta} u)$ is nonzero and its information quadratic form is $u^T J u > 0$. This proves $J > 0$. Direct covariance expansion gives

$$\text{Cov}_0(s_{\text{eff}}, s_\eta) = I_{\vartheta\eta} - I_{\vartheta\eta} I_{\eta\eta}^{-1} I_{\eta\eta} = 0, \qquad \text{Var}_0(s_{\text{eff}}) = J.$$

Remark D.2 (Intuition). A nuisance change can imitate part of a target change. Project the target score onto the nuisance score and remove that projection. Here orthogonality means zero covariance, as verified above. The remaining score has covariance J and carries target information that nuisance motion cannot explain. The dot in $I_{\vartheta \cdot \eta}$ denotes this information after adjusting for η.

Theorem D.3 (Rao test with estimated nuisance)**.** *Assume Theorem 3.4. Let $\widetilde{\eta}_n \xrightarrow{\mathbb{P}} \eta_0$ under $P_{n,0}$ and suppose it solves $U_\eta(\vartheta_0, \widetilde{\eta}_n) = 0$ in the interior with probability tending to one. Suppose a symmetric estimate $\widehat{J}_n$ at the restricted fit $\widetilde{\theta}_n = (\vartheta_0, \widetilde{\eta}_n)$ satisfies $\widehat{J}_n \xrightarrow{\mathbb{P}} J$ under $P_{n,0}$. Then*

$$n^{-1/2} U_\vartheta(\widetilde{\theta}_n) = \Delta_{\vartheta n} - I_{\vartheta\eta} I_{\eta\eta}^{-1} \Delta_{\eta n} + o_{\mathbb{P}}(1) \tag{D.2}$$

under the null and every fixed local alternative. For $\theta_n = \theta_0 + (h_\vartheta, h_\eta)/\sqrt{n}$,

$$S_n = \frac{1}{n} U_\vartheta(\widetilde{\theta}_n)^T \widehat{J}_n^{-1} U_\vartheta(\widetilde{\theta}_n) \rightsquigarrow \chi^2_r(h_\vartheta^T J h_\vartheta).$$

Assign any fixed value when the inverse does not exist. The limit is independent of h_η.

A continuous model information $I(\theta)$ evaluated at a consistent restricted fit is one way to obtain $\widehat{J}_n \xrightarrow{\mathbb{P}} J$; empirical information estimates require their own consistency. The theorem assumes that consistency explicitly.

Proof of Theorem D.3. Expand the restricted nuisance root. Let $d_n = \widetilde{\eta}_n - \eta_0$. On an event with probability tending to one, the segment $(\vartheta_0, \eta_0 + t d_n)$ lies in $\mathcal{U}$, and the restricted score equation gives

$$0 = n^{-1} U_\eta(\theta_0) + H_{\eta\eta,n} d_n, \quad H_{\eta\eta,n} = \int_0^1 P_n \nabla^2_{\eta\eta} \ell_{(\vartheta_0, \eta_0 + t d_n)} \, dt.$$

The envelope gives $\|H_{\eta\eta,n} - P_n \nabla^2_{\eta\eta} \ell_{\theta_0}\| \le \|d_n\| P_n M/2 = o_{\mathbb{P}}(1)$. Thus $H_{\eta\eta,n} \xrightarrow{\mathbb{P}} -I_{\eta\eta}$ and

$$\sqrt{n} d_n = -H_{\eta\eta,n}^{-1} \Delta_{\eta n} = I_{\eta\eta}^{-1} \Delta_{\eta n} + o_{\mathbb{P}}(1) = O_{\mathbb{P}}(1).$$

This is the restricted version of the integral argument in Lemma 5.2, with every block now specified.

Expand the target score at that fit. The same integral identity gives

$$n^{-1/2} U_\vartheta(\widetilde{\theta}_n) = \Delta_{\vartheta n} + H_{\vartheta\eta,n} \sqrt{n} d_n,$$

where

$$H_{\vartheta\eta,n} = \int_0^1 P_n \nabla^2_{\vartheta\eta} \ell_{(\vartheta_0, \eta_0 + t d_n)} \, dt \xrightarrow{\mathbb{P}} -I_{\vartheta\eta}.$$

The Hessian LLN identifies the limit. The envelope bounds its change along the segment by $\|d_n\| P_n M/2$, which tends to zero in probability. Multiplying the vanishing matrix error by the tight $\sqrt{n} d_n$ gives $o_{\mathbb{P}}(1)$. Substitute its preceding expansion to obtain (D.2). Contiguity transfers the remainder and $\widehat{J}_n \xrightarrow{\mathbb{P}} J$ to every $P_{n,h}$.

Identify the local mean and covariance. Put $D = (I_r, -I_{\vartheta\eta} I_{\eta\eta}^{-1})$. Block multiplication gives

$$DI = (J, 0), \qquad DID^T = J.$$

By Theorem 3.8, $D\Delta_n \rightsquigarrow N_r(DIh, DID^T) = N_r(Jh_\vartheta, J)$. This explains exactly why the nuisance shift disappears. The score approximation and the inverse-consistency bound from Lemma G.2 reduce S_n to the quadratic form of this vector with J^{-1}, up to $o_{\mathbb{P}}(1)$. Whitening gives mean $J^{1/2} h_\vartheta$ and identity covariance. The squared norm has noncentrality $\|J^{1/2} h_\vartheta\|^2 = h_\vartheta^T J h_\vartheta$. □

Corollary D.4 (Three tests for linear restrictions)**.** *Under the preceding smooth and information-consistency assumptions, suppose the unrestricted and restricted MLEs are consistent interior maximizers of the full and restricted likelihoods. For Wald, use a consistent estimate of the target covariance J^{-1}, such as $\widehat{J}_n^{-1}$. For the restriction $\vartheta = \vartheta_0$, the LR, efficient-score, and efficient Wald statistics differ by $o_{\mathbb{P}}(1)$ under the null and every fixed local alternative, and have common limit $\chi^2_r(h_\vartheta^T J h_\vartheta)$.*

Proof of Corollary D.4. Write $a = \Delta_{\vartheta n}$, $b = \Delta_{\eta n}$, $F = I_{\vartheta\vartheta}$, $R = I_{\vartheta\eta}$, and $C = I_{\eta\eta}$ within this proof. Let $e = a - RC^{-1}b$ and $J = F - RC^{-1}R^T$. The local quadratic log likelihood for displacement (u, v) is

$$q_n(u, v) = u^T a + v^T b - \tfrac{1}{2}u^T F u - u^T R v - \tfrac{1}{2}v^T C v.$$

Completing the nuisance square gives the exact algebraic identity

$$\begin{aligned} q_n(u, v) = {} & \tfrac{1}{2}b^T C^{-1} b + u^T e - \tfrac{1}{2}u^T J u \\ & - \tfrac{1}{2}\{v - C^{-1}(b - R^T u)\}^T C \{v - C^{-1}(b - R^T u)\}. \end{aligned}$$

For fixed u, it is maximized at $v = C^{-1}(b - R^T u)$. Under $u = 0$ the maximum is $b^T C^{-1} b/2$. Without the restriction, the remaining quadratic in u is maximized at $u = J^{-1}e$, adding $e^T J^{-1} e/2$. Hence twice the difference of the two maximized quadratic values is $e^T J^{-1} e$.

Lemma 5.2 and the restricted-root calculation in Theorem D.3 show that the actual local MLEs equal these quadratic maximizers plus $o_{\mathbb{P}}(1)$ and are tight. The bounded-set uniform LAN expansion can therefore be evaluated at both fits by the explicit localization inequality in the proof of Theorem 5.3. A quadratic evaluated at a point $o_{\mathbb{P}}(1)$ from its maximizer loses only $o_{\mathbb{P}}(1)$, because its curvature matrices are fixed. The actual LR statistic is thus $e^T J^{-1} e + o_{\mathbb{P}}(1)$.

The target displacement of the unrestricted fit is $J^{-1}e + o_{\mathbb{P}}(1)$. Under the null, e has limiting covariance J, so this target displacement has covariance $J^{-1}JJ^{-1} = J^{-1}$. A consistent efficient Wald covariance estimate consequently gives the same quadratic form $e^T J^{-1} e + o_{\mathbb{P}}(1)$. Theorem D.3 gives that expression for the efficient-score test and supplies its local limit. Contiguity transfers the LR and Wald remainder identities as well. □

D.3 Worked example: normal mean with unknown variance

Let $X_i \overset{\text{iid}}{\sim} N(\mu, v)$, $v = \sigma^2 > 0$, and test $\mu = \mu_0$. The per-observation log density and scores are

$$\begin{aligned} \ell_{\mu,v}(x) &= -\tfrac{1}{2}\log(2\pi v) - \frac{(x-\mu)^2}{2v}, \\ s_\mu(x) &= (x-\mu)/v, \qquad s_v(x) = -1/(2v) + (x-\mu)^2/(2v^2). \end{aligned}$$

With $\varepsilon = X - \mu$, the normal moments $\mathbb{E}\varepsilon^2 = v$, $\mathbb{E}\varepsilon^3 = 0$, $\mathbb{E}\varepsilon^4 = 3v^2$ give

$$I_{\mu\mu} = v/v^2 = 1/v, \quad I_{\mu v} = -\mathbb{E}\varepsilon/(2v^2) + \mathbb{E}\varepsilon^3/(2v^3) = 0,$$

$$I_{vv} = \frac{1}{4v^2} - \frac{\mathbb{E}\varepsilon^2}{2v^3} + \frac{\mathbb{E}\varepsilon^4}{4v^4} = \frac{1}{2v^2}.$$

Thus $I_{\mu\cdot v} = 1/v$; the nuisance score is orthogonal to the target score.

The unrestricted and restricted fits are

$$\widehat{\mu} = \overline{X}_n, \quad \widehat{v} = \frac{1}{n}\sum_i (X_i - \overline{X}_n)^2, \quad \widetilde{v} = \frac{1}{n}\sum_i (X_i - \mu_0)^2.$$

Expanding $X_i - \mu_0 = (X_i - \overline{X}_n) + (\overline{X}_n - \mu_0)$ and summing makes the cross term zero. Thus $\widetilde{v} = \widehat{v} + D_n$, where $D_n = (\overline{X}_n - \mu_0)^2$. The three statistics are

$$W_n = nD_n/\widehat{v}, \qquad S_n = nD_n/\widetilde{v}, \qquad \Lambda_n = n\log(\widetilde{v}/\widehat{v}).$$

For the score formula, $U_\mu(\mu_0, \widetilde{v}) = n(\overline{X}_n - \mu_0)/\widetilde{v}$ and null information is $n/\widetilde{v}$, which gives S_n by direct substitution. At either likelihood maximizer the fitted residual sum divided by the fitted variance is n, so subtracting their maximized log likelihoods gives the displayed Λ_n.

Under $\mu_n = \mu_0 + h/\sqrt{n}$, write $X_i = \mu_n + \varepsilon_i$. Then $\sqrt{n}(\overline{X}_n - \mu_0) = h + n^{-1/2}\sum_i \varepsilon_i \rightsquigarrow N(h, v)$, while $\widehat{v} = P_n\varepsilon^2 - \overline{\varepsilon}_n^2 \xrightarrow{\mathbb{P}} v$ and $\widetilde{v} = \widehat{v} + D_n \xrightarrow{\mathbb{P}} v$. In particular, $D_n = O_{\mathbb{P}}(n^{-1})$. On the event of positive fitted variances,

$$W_n - S_n = \frac{nD_n^2}{\widehat{v}\widetilde{v}} = O_{\mathbb{P}}(n^{-1}) = o_{\mathbb{P}}(1).$$

Set $x_n = D_n/\widehat{v} \geq 0$. The exact integral bound $0 \leq x - \log(1+x) = \int_0^x t/(1+t)\,dt \leq x^2/2$ gives

$$|\Lambda_n - W_n| \leq \tfrac{1}{2}nx_n^2 = O_{\mathbb{P}}(n^{-1}) = o_{\mathbb{P}}(1).$$

Thus all three have limit $\chi^2_1(h^2/v)$. Restricted nuisance estimation enters Rao's statistic because its slope is evaluated at the null fit.

For comparison, $s_n^2 = (n-1)^{-1}\sum_i (X_i - \overline{X}_n)^2$ gives the familiar $t_n = \sqrt{n}(\overline{X}_n - \mu_0)/s_n$. Under the normal null, t_n has the exact Student t_{n-1} law. For an integer $\nu \geq 1$, t_ν is the law of $Z/\sqrt{B/\nu}$ for independent $Z \sim N(0,1)$ and $B \sim \chi^2_\nu$. Since $\widehat{v} = (n-1)s_n^2/n$, the likelihood Wald statistic is $W_n = nt_n^2/(n-1)$, showing the finite-sample distinction despite their common limit.

Exercise 4 (Two restrictions)**.** Suppose $\sqrt{n}(\widehat{\theta}_n - \theta_0) \rightsquigarrow N_3(h, I_3)$ locally and test $\theta_1 = \theta_2 = 0$ at $\theta_0 = (0, 0, \theta_{03})$. With known asymptotic covariance, find the Wald statistic and its local limit.

Solution to Exercise 4. Take $G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ and $V = I_3$. Then $GVG^T = I_2$, so $W_n = n(\widehat{\theta}^2_{n1} + \widehat{\theta}^2_{n2})$. Its limit is $\chi^2_2(h_1^2 + h_2^2)$ by Theorem D.1. The third coordinate is a nuisance direction with zero contribution to Gh. □

Exercise 5 (Information lost to a nuisance parameter)**.** For a scalar target and scalar nuisance with information $I = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$, find the efficient score, efficient information, and local noncentrality.

Solution to Exercise 5. Formula (D.1) gives $s_{\text{eff}} = s_\vartheta - \frac{1}{2}s_\eta$ and $J = 4 - 1(1/2)1 = 7/2$. The noncentrality in Theorem D.3 is $(7/2)h_\vartheta^2$. The known-nuisance information would be 4; the projection removes $1/2$ from it. □

E Gaussian half-space testing

Proposition 4.6 bounds one-sided power for a scalar Gaussian mean. Here the null $\{h : a^T h \leq 0\}$ is a half-space, the region on one side of a hyperplane, and all other mean directions are unrestricted. For each alternative we construct a boundary null mean against which the same scalar Neyman–Pearson argument applies. The result extends the bound from Section 4 to a linear target with nuisance directions.

Proposition E.1 (Gaussian half-space test). *Let $Y \sim N_d(h, \Sigma)$ with known $\Sigma > 0$ and nonzero $a \in \mathbb{R}^d$. For $a^T h \le 0$ against $a^T h > 0$, the test*

$$\phi^*_\infty(Y) = \mathbf{1}\left\{ \frac{a^T Y}{\sqrt{a^T \Sigma a}} > z_{1-\alpha} \right\}$$

is UMP of level α. Its power is $1 - \Phi(z_{1-\alpha} - a^T h / \sqrt{a^T \Sigma a})$.

Proof of Proposition E.1. Since $a^T Y \sim N(a^T h, a^T \Sigma a)$, the displayed power follows directly and is at most α for every null mean, with equality on $a^T h = 0$.

Fix an alternative h with $a^T h > 0$. Set $v = a^T \Sigma a > 0$, $c_h = a^T h / v > 0$, and $h_0 = h - \Sigma a c_h$. Then $a^T h_0 = 0$. The log likelihood ratio relative to this null mean is

$$\log \frac{dN_d(h, \Sigma)}{dN_d(h_0, \Sigma)}(y) = (h - h_0)^T \Sigma^{-1} y - \tfrac{1}{2}(h^T \Sigma^{-1} h - h_0^T \Sigma^{-1} h_0) = c_h a^T y - \frac{(a^T h)^2}{2v}.$$

It is strictly increasing in $a^T y$. Under h_0, $a^T Y$ is centered normal with variance v, so the proposed upper-tail region has size exactly α at h_0. Let $L(y)$ be the exponential of the displayed log ratio and put $c = \exp\{c_h \sqrt{v}\, z_{1-\alpha} - (a^T h)^2/(2v)\}$. Then $\phi^*_\infty = \mathbf{1}\{L > c\}$. For any competing composite-null test ϕ, $\mathbb{E}_{h_0} \phi \le \alpha$. Integrating the pointwise inequality $(\phi - \phi^*_\infty)(L - c) \le 0$ as in Proposition 4.6 gives $\mathbb{E}_h \phi \le \mathbb{E}_h \phi^*_\infty$. Since the construction works for every alternative h, the test is UMP. □

For a vector LAN experiment, set $\Sigma = I^{-1}$. Consider competing sequences with local level at each fixed h_0 satisfying $a^T h_0 \le 0$. At any fixed alternative h, choose a subsequence attaining its limsup power and use Theorem 4.1. The common Gaussian limit test has level at every such h_0, so Proposition E.1 bounds the chosen limsup. The bound is attained by rejection for large values of

$$\frac{a^T I^{-1} \Delta_n}{\sqrt{a^T I^{-1} a}},$$

since Theorem 3.8 makes this statistic asymptotically $N(a^T h / \sqrt{a^T I^{-1} a}, 1)$ under each $P_{n,h}$, and Lemma 1.6(ii) gives its limiting rejection probability.

F An influence-function explanation of the information bound

The main argument bounds the power of every test. This appendix explains why the same information also bounds the variance of a locally regular asymptotically linear estimator. It closes the connection between the local slopes in Section 1 and the sample-size ratios in Section 2. Local regularity below means that centering at the local true parameter gives the same limiting law for every fixed local shift h.

Proposition F.1 (Regular influence functions)**.** *In a scalar LAN model with central sequence $n^{-1/2}\sum_i s_0(X_i)$, assume $\mathbb{E}_0 s_0 = 0$ and $\mathbb{E}_0 s_0^2 = I \in (0,\infty)$. Suppose under $P_{n,0}$ that*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}}\sum_i \varphi(X_i) + o_{\mathbb{P}}(1), \quad \mathbb{E}_0\varphi = 0, \quad \mathbb{E}_0\varphi^2 = V \in (0,\infty).$$

Assume local regularity: under every fixed $P_{n,h}$, $\sqrt{n}(\widehat{\theta}_n - \theta_n(h)) \rightsquigarrow N(0,V)$. Then $\mathbb{E}_0[\varphi s_0] = 1$, $V \geq I^{-1}$, and equality holds exactly when $\varphi = I^{-1}s_0$ almost surely. A consistently studentized one-sided test based on this estimator has slope $1/\sqrt{V} \leq \sqrt{I}$.

Proof of Proposition F.1. Let $c = \mathbb{E}_0[\varphi s_0]$, which is finite by Cauchy–Schwarz. The joint iid CLT gives

$$\left(\frac{1}{\sqrt{n}}\sum_i \varphi(X_i), \Delta_n\right) \rightsquigarrow (Z_\varphi, Z_s) \sim N_2\left(0, \begin{pmatrix} V & c \\ c & I \end{pmatrix}\right).$$

This covariance matrix may be singular; the following moment-generating-function calculation remains valid. Under the likelihood tilt $e^{hZ_s - h^2 I/2}$,

$$\begin{aligned}\mathbb{E}[e^{tZ_\varphi}e^{hZ_s - h^2 I/2}] &= \exp\{\tfrac{1}{2}(Vt^2 + 2cth + Ih^2) - \tfrac{1}{2}Ih^2\} \\ &= \exp\{cth + \tfrac{1}{2}Vt^2\}.\end{aligned}$$

This is the moment-generating function of $N(ch, V)$. LAN gives joint convergence with the likelihood ratio, whose limit has mean one. Lemma 3.7 therefore shows that the influence-function sum converges under $P_{n,h}$ to $N(ch, V)$. Contiguity transfers the expansion error, so after subtracting h,

$$\sqrt{n}(\widehat{\theta}_n - \theta_n(h)) \rightsquigarrow N((c-1)h, V).$$

Local regularity requires this law to equal $N(0,V)$ for every h. Taking any nonzero h forces $c = 1$.

Cauchy–Schwarz now gives $1 = c^2 \leq VI$. More explicitly,

$$\mathbb{E}_0(\varphi - I^{-1}s_0)^2 = V - 2I^{-1}c + I^{-2}I = V - I^{-1} \geq 0.$$

Equality holds exactly when this nonnegative square is zero almost surely, proving the equality characterization. Corollary 1.9 gives slope $1/\sqrt{V}$, which is at most $\sqrt{I}$ by the variance bound. □

Remark F.2 (Intuition). Local regularity requires the estimator to respond to a small parameter change with the correct first-order displacement. The covariance identity $\mathbb{E}_0[\varphi s_0] = 1$ expresses that response. Its variance cost is at least I^{-1}, giving the same maximal slope as the testing envelope.

G Probability and matrix tools

These results supply the shared steps behind studentization, likelihood tilting, matrix standardization, and the sign-test limits. Lemma G.1 combines distributional limits with small errors; Lemma G.2 handles differentiable maps and inverses; Lemma G.3 permits the sign probability to vary with sample size. Theorem G.4 supplies the density used to construct the Gaussian test in Theorem 4.1. Write Q_n for the probability measure governing the random variables at index n.

Lemma G.1 (Tightness, continuous mappings, and small remainders)**.** *Suppose $X_n \rightsquigarrow X$ in a finite-dimensional Euclidean space.*

1. *$X_n = O_{\mathbb{P}}(1)$. If f is continuous, then $f(X_n) \rightsquigarrow f(X)$. If $Y_n \xrightarrow{\mathbb{P}} y$ for a constant y and f is continuous at y, then $f(Y_n) \xrightarrow{\mathbb{P}} f(y)$.*
2. *If $R_n = o_{\mathbb{P}}(1)$ and $Y_n = O_{\mathbb{P}}(1)$, then $\|Y_n\|\|R_n\| = o_{\mathbb{P}}(1)$.*
3. *If $R_n = o_{\mathbb{P}}(1)$, then $(X_n, R_n) \rightsquigarrow (X, 0)$. In particular, $X_n + R_n \rightsquigarrow X$ when the dimensions agree. More generally, if $Y_n \xrightarrow{\mathbb{P}} y$ and f is continuous, then $f(X_n, Y_n) \rightsquigarrow f(X, y)$.*

Proof of Lemma G.1. For tightness, apply the closed-set bound in Lemma 1.6(i) to $\{x : \|x\| \geq M\}$. Its limiting probability decreases to zero as $M \to \infty$, giving $X_n = O_{\mathbb{P}}(1)$. For a bounded continuous g, the composition $g \circ f$ is bounded and continuous, so $\mathbb{E}g(f(X_n)) \to \mathbb{E}g(f(X))$. By the definition of convergence in distribution in Section 0, this proves $f(X_n) \rightsquigarrow f(X)$.

For the probability-convergence assertion, fix $\epsilon > 0$. Continuity at y gives a $\delta > 0$ such that $\|u - y\| < \delta$ implies $\|f(u) - f(y)\| < \epsilon$. Hence

$$\mathsf{Q}_n\big(\|f(Y_n) - f(y)\| > \epsilon\big) \leq \mathsf{Q}_n\big(\|Y_n - y\| \geq \delta\big) \longrightarrow 0.$$

This is $f(Y_n) \xrightarrow{\mathbb{P}} f(y)$.

For the product, for every $M, \epsilon > 0$,

$$\mathsf{Q}_n(\|Y_n\|\|R_n\| > \epsilon) \leq \mathsf{Q}_n(\|Y_n\| > M) + \mathsf{Q}_n(\|R_n\| > \epsilon/M).$$

First choose M so the first term has arbitrarily small limsup, then let $n \to \infty$ in the second term.

For the joint limit, let $g(x, r)$ be bounded and continuous. Given $\epsilon > 0$, tightness gives a finite M for which $\limsup_n \mathsf{Q}_n(\|X_n\| > M) < \epsilon$. Uniform continuity of g on the compact set $\{\|x\| \leq M, \|r\| \leq 1\}$ gives a $\delta \in (0, 1)$ such that $|g(x, r) - g(x, 0)| < \epsilon$ there whenever $\|r\| < \delta$. Hence

$$\big|\mathbb{E}g(X_n, R_n) - \mathbb{E}g(X_n, 0)\big| \leq \epsilon + 2\|g\|_\infty\{\mathsf{Q}_n(\|X_n\| > M) + \mathsf{Q}_n(\|R_n\| \geq \delta)\}.$$

Its limsup is at most $\epsilon + 2\|g\|_\infty\epsilon$. Let $\epsilon \downarrow 0$ and use $\mathbb{E}g(X_n, 0) \to \mathbb{E}g(X, 0)$. Addition is a continuous map. For a nonzero constant y, use $R_n = Y_n - y$ and then the continuous map $(x, r) \mapsto f(x, y + r)$. □

Lemma G.2 (Differentiable maps and matrix inversion)**.** *Let $a_n \to \infty$, $a_n(\widehat{\theta}_n - \theta_0) = O_{\mathbb{P}}(1)$, and let g be differentiable at θ_0 with derivative G. Then*

$$a_n\{g(\widehat{\theta}_n) - g(\theta_0)\} = Ga_n(\widehat{\theta}_n - \theta_0) + o_{\mathbb{P}}(1).$$

If matrices $C_n \xrightarrow{\mathbb{P}} C$ with nonsingular C, then C_n is nonsingular with probability tending to one and $C_n^{-1} \xrightarrow{\mathbb{P}} C^{-1}$ on that event. If C and C_n are symmetric and $C > 0$, then $C_n > 0$ with probability tending to one.

Proof of Lemma G.2. Differentiability gives $g(\theta_0 + t) - g(\theta_0) = Gt + r(t)$, where $\|r(t)\|/\|t\| \to 0$ as $t \to 0$; set this ratio to zero at $t = 0$. Tightness and $a_n \to \infty$ imply $\widehat{\theta}_n - \theta_0 \xrightarrow{\mathbb{P}} 0$. Consequently

$$a_n\|r(\widehat{\theta}_n - \theta_0)\| = \{a_n\|\widehat{\theta}_n - \theta_0\|\}\frac{\|r(\widehat{\theta}_n - \theta_0)\|}{\|\widehat{\theta}_n - \theta_0\|} = O_{\mathbb{P}}(1)o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1).$$

This proves the first assertion.

For $r \times r$ matrices, write $C_n = C(I_r + D_n)$ with $D_n = C^{-1}(C_n - C) \xrightarrow{\mathbb{P}} 0$. On $\|D_n\| < 1/2$, the matrix geometric series, also called the Neumann series, gives $(I_r + D_n)^{-1} = \sum_{j=0}^{\infty}(-D_n)^j$ and norm at most two. Thus

$$\|C_n^{-1} - C^{-1}\| \le 2\|C^{-1}\|^2\|C_n - C\| \xrightarrow{\mathbb{P}} 0.$$

The event has probability tending to one. In the symmetric positive definite case, let $\lambda_{\min}(C) > 0$ be the smallest eigenvalue of C. For every unit vector u, $u^T C_n u \ge \lambda_{\min}(C) - \|C_n - C\|$. This is positive on $\|C_n - C\| < \lambda_{\min}(C)/2$, which proves the final assertion. □

A triangular array has one row of variables for each sample size; its row distribution may change with n. To prove its limit below, we use the characteristic function $t \mapsto \mathbb{E}e^{itY}$ of a real variable Y, with $i^2 = -1$. The characteristic-function continuity theorem states that pointwise convergence of these functions to the characteristic function of a law implies convergence in distribution to that law.

Lemma G.3 (Bernoulli triangular-array CLT)**.** *For each n, let $B_{n1}, \dots, B_{nn}$ be independent Bernoulli variables with common parameter p_n. Suppose $p_n \to p \in (0,1)$. For all sufficiently large n, define*

$$Z_n = \frac{\sum_{i=1}^{n} B_{ni} - np_n}{\sqrt{np_n(1-p_n)}}.$$

Then $Z_n \rightsquigarrow N(0,1)$. Independence is required only within each row, not across rows.

Proof of Lemma G.3. Put $\xi_{ni} = (B_{ni} - p_n)/\sqrt{np_n(1-p_n)}$. For all sufficiently large n, $p_n(1-p_n)$ is bounded below by a positive constant. Thus $|\xi_{ni}| \le C/\sqrt{n}$ for some fixed $C < \infty$, $\mathbb{E}\xi_{ni} = 0$, and $\mathbb{E}\xi_{ni}^2 = 1/n$. Taylor's formula for the complex exponential gives, for each fixed real t,

$$\mathbb{E}e^{it\xi_{n1}} = 1 - \frac{t^2}{2n} + r_n(t), \qquad |r_n(t)| \le \frac{|t|^3}{6}\mathbb{E}|\xi_{n1}|^3 = O(n^{-3/2}).$$

Independence within a row makes the characteristic function of the sum equal to the nth power of this expression. Since $n\log(1 - t^2/(2n) + O(n^{-3/2})) \to -t^2/2$, it converges to $e^{-t^2/2}$, the characteristic function of $N(0,1)$. The characteristic-function continuity theorem gives the stated distributional limit. □

The following standard theorem extends the notion of a density to two measures on the same measurable space.

Theorem G.4 (Radon–Nikodym theorem for finite measures)**.** *Let m and ν be finite nonnegative measures on a measurable space $(S, \mathcal{A})$. Suppose m is absolutely continuous with respect to ν, meaning that $\nu(B) = 0$ implies $m(B) = 0$ for every $B \in \mathcal{A}$. Then there is a nonnegative, ν-integrable measurable function r, unique up to a ν-null set, such that*

$$m(B) = \int_B r \, d\nu \qquad \text{for every } B \in \mathcal{A}.$$

The function r is called the Radon–Nikodym derivative, or density of m relative to ν, and is written $r = dm/d\nu$. If also $m(B) \le \nu(B)$ for every $B \in \mathcal{A}$, then $0 \le r \le 1$ ν-almost everywhere.

The following exercise verifies the variance used in Example 2.6.

Exercise 6 (Variance of logistic errors)**.** Let U be uniform on $(0,1)$, let $s > 0$, and set $X = s\log\{U/(1-U)\}$. Show that X has cdf $F(x) = (1+e^{-x/s})^{-1}$, mean zero, and variance $\pi^2 s^2/3$. You may use $\sum_{k=1}^{\infty} k^{-2} = \pi^2/6$.

Solution to Exercise 6. The transformation is increasing, so $\mathbb{P}(X \le x) = \mathbb{P}\{U \le (1+e^{-x/s})^{-1}\} = (1+e^{-x/s})^{-1}$. Since $\int_0^1 |\log u|\, du = 1$, X is integrable. Replacing U by $1-U$ changes the sign of X without changing its distribution, so $\mathbb{E}X = 0$.

Substitution $u = e^{-t}$ and two integrations by parts give

$$\int_0^1 \log^2 u \, du = \int_0^\infty t^2 e^{-t}\, dt = 2.$$

To calculate the cross term, expand $-\log(1-u) = \sum_{k=1}^{\infty} u^k/k$. All terms after multiplication by $-\log u$ are nonnegative, so monotone convergence gives

$$\int_0^1 \log u \log(1-u)\, du = \sum_{k=1}^{\infty} \frac{1}{k} \int_0^1 u^k(-\log u)\, du = \sum_{k=1}^{\infty} \frac{1}{k(k+1)^2}.$$

The last integral equals $1/(k+1)^2$ by integration by parts. The identity

$$\frac{1}{k(k+1)^2} = \frac{1}{k} - \frac{1}{k+1} - \frac{1}{(k+1)^2}$$

then gives the sum $1 - (\pi^2/6 - 1) = 2 - \pi^2/6$. By symmetry, the integral of $\log^2(1-u)$ also equals two. Hence

$$\begin{aligned}\mathrm{Var}(X) &= s^2 \int_0^1 \{\log u - \log(1-u)\}^2\, du \\ &= s^2\{2 + 2 - 2(2 - \pi^2/6)\} = \frac{\pi^2 s^2}{3}.\end{aligned}$$

□

H Minimum sample sizes and local efficiency

Section 2 interprets relative efficiency as a comparison of the sample sizes needed to attain a common power. This appendix justifies that interpretation when each sample size is the smallest integer attaining the target. Theorem 2.2 assumes calibrated sequences with the required local scaling; here we derive that scaling from the tests' power functions. Because the local parameter $h = \sqrt{n}\delta$ varies as we search over sample sizes, the fixed-h limits in Section 1 must first be strengthened to uniform power approximations. We then invert those approximations and specify how to exclude a fixed small sample size whose power already exceeds the target.

H.1 Uniform local power for location tests

Fix the level $\alpha \in (0,1)$ and set $\theta_0 = 0$. The labels mean and sign identify the two tests in Section 1.4; they are not mathematical operations. Their finite-sample power functions are

$$\begin{aligned}\beta_n^{\mathrm{mean}}(\delta) &= P_{\theta=\delta}\big(T_n^{\mathrm{mean}} > z_{1-\alpha}\big), \\ \beta_n^{\mathrm{sign}}(\delta) &= P_{\theta=\delta}\big(T_n^{\mathrm{sign}} > z_{1-\alpha}\big).\end{aligned}$$

Thus $\beta_n^{\text{mean}}(\delta)$ is the probability that the mean test rejects when the true location is δ, based on n observations; $\beta_n^{\text{sign}}(\delta)$ has the same meaning for the sign test. The mean statistic is defined as in Proposition 1.11, including its value at zero sample variance.

For a local shift $\delta = h/\sqrt{n}$, the limiting power functions and local slopes are

$$\begin{aligned} b_{\text{mean}}(h) &= 1 - \Phi(z_{1-\alpha} - h/\sigma), & \kappa_{\text{mean}} &= 1/\sigma, \\ b_{\text{sign}}(h) &= 1 - \Phi(z_{1-\alpha} - 2f(0)h), & \kappa_{\text{sign}} &= 2f(0). \end{aligned}$$

For $j \in \{\text{mean}, \text{sign}\}$, both functions have the form $b_j(h) = 1 - \Phi(z_{1-\alpha} - \kappa_j h)$. Here κ_j is the coefficient of h in the limiting mean of $T_n^j \rightsquigarrow N(\kappa_j h, 1)$ under the local alternative; it is not the derivative of b_j. The distinction is that $\beta_n^j(h/\sqrt{n})$ is the power at sample size n, whereas $b_j(h)$ is its limit.

For the exactly calibrated sign test (A.1), define $\beta_n^{\text{exact sign}}(\delta) = \mathbb{E}_{\theta=\delta}\phi_n$ and $b_{\text{exact sign}}(h) = b_{\text{sign}}(h)$. In the proof below, j labels any of these three tests.

Proposition H.1 (Uniform local power on bounded intervals)**.** *Under the assumptions of Proposition 1.11, for every finite $H > 0$,*

$$\sup_{0 \le h \le H} \left| \beta_n^{\text{mean}}(h/\sqrt{n}) - b_{\text{mean}}(h) \right| \longrightarrow 0, \qquad \kappa_{\text{mean}} = 1/\sigma. \tag{H.1}$$

Under the assumptions of Proposition 1.12, the same conclusion holds for both the normal-cutoff and the exactly calibrated sign tests, with $\kappa_{\text{sign}} = 2f(0)$.

Proof of Proposition H.1. First let $n_k \to \infty$ and $h_k \to h$ in $[0, H]$. We prove convergence of the powers at $\theta_k = h_k/\sqrt{n_k}$.

For the mean statistic, translation leaves the estimated variance unchanged. Under θ_k, its numerator is

$$\sqrt{n_k}\overline{X}_{n_k} = h_k + \frac{1}{\sqrt{n_k}} \sum_{i=1}^{n_k} \varepsilon_i.$$

The CLT gives convergence of the sum on the right to $N(0, \sigma^2)$, and the LLN gives $\widehat{\sigma}_{n_k} \xrightarrow{\mathbb{P}} \sigma > 0$. In particular, $P_{n_k, h_k}(\widehat{\sigma}_{n_k} = 0) \to 0$. By Lemma G.1 and the probability comparison in the proof of Corollary 1.9,

$$T_{n_k}^{\text{mean}} \rightsquigarrow N(h/\sigma, 1).$$

Apply Lemma 1.6(ii) with $D = (z_{1-\alpha}, \infty)$. The limiting normal law has no atom at the endpoint, giving $\beta_{n_k}^{\text{mean}}(h_k/\sqrt{n_k}) \to b_{\text{mean}}(h)$.

For either sign test, the Bernoulli success probability is

$$p_k = F(h_k/\sqrt{n_k}), \qquad \sqrt{n_k}(p_k - 1/2) = f(0)h_k + o(1) \longrightarrow f(0)h.$$

To justify the remainder, write $F(t) = 1/2 + f(0)t + tr(t)$ with $r(t) \to 0$ as $t \to 0$, and set $r(0) = 0$. Its contribution is $h_k r(h_k/\sqrt{n_k}) \to 0$, since h_k is bounded. Lemma G.3 and the decomposition in the proof of Proposition 1.12 give

$$T_{n_k}^{\text{sign}} \rightsquigarrow N(2f(0)h, 1).$$

Lemma 1.6(ii) therefore gives convergence of the normal-cutoff power to $b_{\text{sign}}(h)$.

For the exactly calibrated test, Proposition A.1 proves that the standardized cutoff $t_n = 2(k_n - n/2)/\sqrt{n}$ tends to $z_{1-\alpha}$. Hence, by Lemma G.1,

$$T_{n_k}^{\text{sign}} - t_{n_k} \rightsquigarrow N(2f(0)h - z_{1-\alpha}, 1).$$

Apply Lemma 1.6(ii) to $(0, \infty)$ for the strict rejection part. Apply its closed-set bound to $\{0\}$ for the boundary part:

$$\limsup_{k\to\infty} P_{n_k,h_k}(K_{n_k} = k_{n_k}) \le \mathbb{P}\{N(2f(0)h - z_{1-\alpha}, 1) = 0\} = 0.$$

The randomized boundary contribution is bounded by this probability. Thus the exactly calibrated power also converges to $b_{\text{sign}}(h)$.

Finally, fix one of the three tests and suppose that its uniform assertion fails. There then exist $\epsilon > 0$, indices $n_k \to \infty$, and $h_k \in [0, H]$ for which the absolute approximation error is at least ϵ. Compactness gives a subsequence with $h_k \to h \in [0, H]$. The sequential limits just proved give $\beta_{n_k}^j(h_k/\sqrt{n_k}) \to b_j(h)$, while continuity gives $b_j(h_k) \to b_j(h)$. Their difference tends to zero, contradicting the choice of ϵ. □

H.2 Inverting the power approximation

For a general sequence of tests, write $\beta_n(\delta)$ for power at $\theta_0 + \delta$, where $\delta \ge 0$. Fix a target $q \in (\alpha, 1)$ and an integer $n_{\min} \ge 1$. Define

$$n_*(\delta) = \min\{n \ge n_{\min} : \beta_n(\delta) \ge q\}. \tag{H.2}$$

Initially the set may be empty; the next theorem proves that it is nonempty for all sufficiently small positive δ. The lower cutoff $n_{\min}$ specifies the sample sizes at which the test is used.

Theorem H.2 (Minimum sample size from uniform local power)**.** *Suppose $b : [0, \infty) \to [0, 1]$ is continuous and strictly increasing, with $b(0) = \alpha$ and $b(h) \to 1$ as $h \to \infty$. Assume:*

(i) For every finite $H > 0$,

$$\sup_{0\le h\le H} |\beta_n(h/\sqrt{n}) - b(h)| \longrightarrow 0.$$

(ii) For each fixed integer $n \ge n_{\min}$,

$$\limsup_{\delta\downarrow 0} \beta_n(\delta) < q.$$

Let $h_* > 0$ be the unique solution of $b(h_*) = q$. Then, as $\delta \downarrow 0$,

$$\delta^2 n_*(\delta) \longrightarrow h_*^2, \qquad \beta_{n_*(\delta)}(\delta) \longrightarrow q. \tag{H.3}$$

Moreover, assumption (i) alone ensures that assumption (ii) holds for every sufficiently large fixed choice of $n_{\min}$.

Proof of Theorem H.2. Continuity, strict increase, and the endpoint values give a unique $h_* > 0$ with $b(h_*) = q$. Fix $\epsilon \in (0, h_*)$, and set $a = h_* - \epsilon$ and $c = h_* + \epsilon$. Then $b(a) < q < b(c)$.

Upper bound and existence. Let $m(\delta) = \lceil c^2/\delta^2 \rceil$, where $\lceil x \rceil$ is the smallest integer at least x. As $\delta \downarrow 0$, $m(\delta) \to \infty$ and $\sqrt{m(\delta)}\delta \to c$. Choose a fixed $H > c$. For all sufficiently small δ, the local

parameter $\sqrt{m(\delta)}\delta$ lies in $[0, H]$. Assumption (i), evaluated at this sample size and local parameter, and continuity of b imply

$$\beta_{m(\delta)}(\delta) = b(\sqrt{m(\delta)}\delta) + o(1) \longrightarrow b(c) > q.$$

Thus $m(\delta)$ belongs to the set in (H.2) eventually, and

$$\limsup_{\delta \downarrow 0} \delta^2 n_*(\delta) \leq \lim_{\delta \downarrow 0} \delta^2 m(\delta) = c^2.$$

Lower bound. Set $\gamma = (q - b(a))/2 > 0$. Assumption (i) with $H = a$ gives an integer $N > n_{\min}$ such that, for every $n \geq N$ and $0 \leq h \leq a$,

$$\beta_n(h/\sqrt{n}) \leq b(h) + \gamma \leq b(a) + \gamma < q.$$

Consequently, no $n \geq N$ with $n\delta^2 \leq a^2$ can attain the target. For each of the finitely many integers $n_{\min} \leq n < N$, assumption (ii) supplies a positive neighborhood of zero in which $\beta_n(\delta) < q$. Intersecting these finitely many neighborhoods excludes all such n for sufficiently small δ. Hence $n_*(\delta)\delta^2 > a^2$ eventually, and

$$\liminf_{\delta \downarrow 0} \delta^2 n_*(\delta) \geq a^2.$$

Let $\epsilon \downarrow 0$ in the two bounds to obtain the first limit in (H.3). In particular, $n_*(\delta) \to \infty$ and $\sqrt{n_*(\delta)}\delta \to h_*$. Applying (i) once more on a fixed interval containing these local parameters gives the second limit.

Choosing the lower cutoff. Choose any $a_0 \in (0, h_*)$ and put $\gamma_0 = (q - b(a_0))/2$. By (i), there is N_0 such that $\beta_n(h/\sqrt{n}) \leq b(a_0) + \gamma_0 < q$ for every $n \geq N_0$ and $h \in [0, a_0]$. For each fixed $n \geq N_0$, take $\delta \downarrow 0$ with $\sqrt{n}\delta \leq a_0$. This yields

$$\limsup_{\delta \downarrow 0} \beta_n(\delta) \leq b(a_0) + \gamma_0 < q.$$

Thus any fixed $n_{\min} \geq N_0$ satisfies (ii). □

Remark H.3 (The role of the assumptions). Assumption (i) controls sample sizes that diverge while the local parameter varies. Assumption (ii) excludes a fixed sample size that already reaches the target near the null. It holds, for example, if every fixed-n power function is continuous at zero and $\beta_n(0) \leq \alpha$. Monotonicity of power in n is not required: the proof excludes every sample size below the lower bound and exhibits one above it. For tests calibrated only asymptotically, the theorem permits a sufficiently large fixed lower cutoff; it does not assert the same conclusion for a minimum taken over all positive integers without condition (ii).

H.3 Consequences for the mean and sign tests

Corollary H.4 (Minimum-sample-size ARE in the location model)**.** *Under the common assumptions for the mean and sign tests, fix $q = 1 - \eta \in (\alpha, 1)$ and put $C = z_{1-\alpha} + z_{1-\eta} > 0$. For each test, define its minimum sample size by (H.2). If its fixed-sample-size powers satisfy assumption (ii) of Theorem H.2, then*

$$\delta^2 n_{\text{mean}}(\delta) \longrightarrow C^2\sigma^2, \qquad \delta^2 n_{\text{sign}}(\delta) \longrightarrow \frac{C^2}{4f(0)^2}.$$

Such a common fixed lower cutoff $n_{\min}$ always exists. Consequently,

$$\frac{n_{\text{mean}}(\delta)}{n_{\text{sign}}(\delta)} \longrightarrow 4\sigma^2 f(0)^2 = \text{ARE}(\text{sign}, \text{mean}).$$

The exact sign test (A.1) has the same limits with $n_{\min} = 1$.

Proof of Corollary H.4. Proposition H.1 verifies assumption (i) for each test. Its limiting power is $b_j(h) = 1 - \Phi(z_{1-\alpha} - \kappa_j h)$, a continuous strictly increasing function with $b_j(0) = \alpha$ and limit one. Solving $b_j(h_j) = 1 - \eta$ gives $h_j = C/\kappa_j$. Theorem H.2 now yields $\delta^2 n_j(\delta) \to C^2/\kappa_j^2$. Substituting $\kappa_{\text{mean}} = 1/\sigma$ and $\kappa_{\text{sign}} = 2f(0)$ proves the two limits. Taking their ratio gives the stated ARE. The final assertion of that theorem supplies a lower cutoff for each test; their maximum is a common cutoff.

For the exactly calibrated sign test, at a fixed n the power is the finite sum

$$\beta_n^{\text{exact sign}}(\delta) = \sum_{k=k_n+1}^{n} \binom{n}{k} F(\delta)^k \{1 - F(\delta)\}^{n-k} + \rho_n \binom{n}{k_n} F(\delta)^{k_n} \{1 - F(\delta)\}^{n-k_n}.$$

Continuity of F at zero makes this power continuous at zero. Exact calibration gives its value $\alpha < q$ there, so assumption (ii) holds for every $n \geq 1$. Theorem H.2 therefore applies with $n_{\min} = 1$. □

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